



PRESIDENCY UNIVERSITY

Private University Estd. in Karnataka State by Act No. 41 of 2013
Itgalpura, Rajankunte, Yelahanka, Bengaluru – 560064



MOBILE APP FOR DIRECT MARKET ACCESS FOR FARMERS

A PROJECT REPORT

Submitted by

DARSHAN B- 20221CSE0735

SUJITH M MADDIN- 20221CSE0738

NAGASHREE M N- 20221CSE0747

Under the guidance of,

Mr. JERRIN JOE FRANCIS

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING

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BONAFIDE CERTIFICATE

Certified that this report “MOBILE APP FOR DIRECT MARKET ACCESS FOR FARMERS” is a bonafide work of “DARSHAN B (20221CSE0735), SUJITH M MADDIN (20221CSE0738), NAGASHREE M N (20221CSE0747)”, who have successfully carried out the project work and submitted the report for partial fulfilment of the requirements for the award of the degree of BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE AND ENGINEERING during 2025-26.

Mr. Jerrin Joe Francis
Project Guide
PSCS
Presidency University

Dr. Jayavadivel Ravi
Program Project
Coordinator PSCS
Presidency University

**Dr. Sampath A K /
Dr. Geetha A**
School Project
Coordinator PSCS
Presidency University

Dr. Asif Mohamed HB
Head of Department
PSCS
Presidency University

Dr. Shakkeera L
Associate Dean
PSCS
Presidency University

Dr. Duraipandian N
Dean
PSCS & PSIS
Presidency University

Examiners

Sl.no	Name	Signature	Date
1.			
2.			



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DECLARATION

We the students of final year B.Tech in COMPUTER SCIENCE AND ENGINEERING at Presidency University, Bengaluru, named DARSHAN B, SUJITH M MADDIN, NAGASHREE M N , hereby declare that the project work titled **“MOBILE APP FOR DIRECT MARKET ACCESS FOR FARMERS”** has been independently carried out by us and submitted in partial fulfillment for the award of the degree of B.Tech in COMPUTER SCIENCE AND ENGINEERING during the academic year of 2025-26. Further, the matter embodied in the project has not been submitted previously by anybody for the award of any Degree or Diploma to any other institution.

DARSHAN B	USN: 20221CSE0735
SUJITH M MADDIN	USN: 20221CSE0738
NAGASHREE M N	USN: 20221CSE0747

PLACE: BENGALURU

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DARSHAN B
SUJITH M MADDIN
NAGASHREE M N

Abstract

Agriculture is a crucial pillar of the rural economy, yet farmers continue to face financial instability due to inefficiencies, exploitation, and lack of transparency in traditional supply chains. Middlemen play a dominant role in price determination, often reducing farmers' earnings and limiting their access to fair markets. To address these challenges, this project introduces a mobile-based platform that connects farmers directly with verified buyers—such as consumers, retailers, and wholesalers—while offering a multilingual interface, voice search, and partial offline accessibility to support users in low-connectivity regions. The system enables farmers to easily register, upload crop details, set base prices, and participate in transparent bidding processes or direct negotiations.

The application integrates secure digital payment options to ensure fast and reliable transactions while minimizing risks associated with cash-based trade. Additional features, including crop advisory insights and logistics coordination, empower farmers to make informed decisions regarding production, pricing, and delivery. Built using Android Studio with Firebase as the backend, the platform provides real-time data synchronization, security, and scalability, ensuring a seamless user experience. Overall, the project aims to promote fair trade, enhance market transparency, and strengthen farmer–buyer relationships, thereby supporting rural economic growth and contributing to the ongoing digital transformation in agriculture.

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CHAPTER-1

INTRODUCTION

1.1 Background

Agriculture is a vital component in the maintenance of rural livelihoods and national economies and small and medium-sized farmers are frequently hindered by the difficulties of fair and profitable market access. The old system of agricultural trade is dominated by intermediaries and middlemen who dictate prices and the distribution meaning that farmers earning little profits in their hard labor. This is also associated with the inefficiency, late payments, and low levels of transparency in transactions which can be further enabled by the development of digital technology and mobile connectivity in the last few years. The mobile-based platforms can empower farmers since they can now sell their produce to the consumers, retailers as well as wholesalers without having to depend on the intermediaries. This project has the potential of maintaining fair trade and enhancing merit and responsibility at the farming marketplace; on the other hand, it seeks to fill this digital void by designing an Android-based application to enable farmers to have direct access to trusted buyers. The platform makes it easy to list crops, bargain prices and to make payments to form an effective, open and easy to transact with digital marketplace.

1.2 Statistics of project

Agriculture remains an important socio-economic sector in India with a high percentage of the population still engaged in it and has been supporting livelihood of the rural people. The Ministry of Agriculture and Farmers Welfare (2023) has observed that in India, about 54.6 percent of rural labour force have relied on agriculture as the leading form of their income. In spite of this participation, more than 86% of Indian farmers are termed as small and marginal who do not own more than two hectares of land (NITI Aayog, 2022). Most farmers cannot get fair prices of their produce due to low bargaining power and limited access to the market and research studies have shown that farmers usually end up with 25%35% of the end-market price, with the rest of the value going to the intermediaries and middlemen. The difference between this margin has sharp implications on the revenues of the farmers, economic balance as well as sustainability of agricultural practices in the long-term. Karnataka has been facing the same predicaments in the agricultural sector especially among

the small and marginal farmers.

The same case applies to the agricultural sector in Karnataka, especially those that are small and marginal farmers. According to the Karnataka Agriculture Market Board (2023), the farmers in the state.

In many cases receive unequal income flows because of price volatility, inaccessibility to remote markets and are reliant on Agricultural Produce Market Committees (APMCs) to sell their products. A study carried out at the state level by the University of Agricultural Sciences, Bengaluru (2023) revealed that proper digital connectivity between farmers and buyers can boost farm revenue between 20 percent and 30 percent due to the ability to realise better prices and the decrease in the participation of middlemen in the relationship. It shows that there is a high demand in the region on the solutions of market access that are technologically based and farmer-friendly.

Digital platforms which enhance the linkage between farmers and markets are also on the increasing interest of the global agricultural landscape. As an example, MFarm platform in Kenya has shown that digital access to market prices and buyer networks can boost farmers income up to 20 percent by making them less exploited and making them more transparent (Kabubo- Mariara, 2013). Other positive effects have been witnessed in the developing world where the digital agriculture systems provide farmers with pricing knowledge, advisory services, direct trading facilities, and the transformative potential of mobile-based interventions on smallholder farming communities is seen.

In order to put the income gap faced by the farmers in accordance with the market structure and the possible re-energization with the help of the direct access platforms, Table 1.1 offers a comparative.

overview:

Market Model	Farmer's Share of consumer Price	Presence of Middlemen	Accessibility	Impact on Farmer Income

Traditional Market System	25% – 35%	High	Moderate to High Dependence on Physical Mandis	Low and Unstable
Digital Advisory Platforms (No Direct Selling)	30% – 40%	Medium	Advisory-Focused, Limited Sales Features	Moderate
Direct Market Access (Digital Platforms)	50% – 80%	Minimal to None	High (Digital Connectivity Required)	High and More Stable
Case Study: Kenya – MFarm	~70%	None	High (Mobile/SMS Based)	Significant Income Growth

Table 1.1: Comparative Overview of Farmer Earnings in Traditional vs Direct Market Access Models

1.3 Prior Existing Technologies

Prior to the creation of this project, a number of digital applications and mobile platforms were already existing to assist agricultural marketing and enhance the connection of farmers. Government programs like eNAM (National Agriculture Market) and commercial programs like AgriBazaar, KisanMandi and AgroStar are offering on-line trade and counseling to farmers. The systems have been very helpful in the promotion of digital trade as well as providing information related to the market. Nevertheless, the majority of these available outlets are mostly limited to large-scale agricultural producers or need high internet connection and digital literacy which is lacking in most rural farmers. Moreover, they tend to use fixed pricing as compared to variable negotiating models an aspect that restricts farmers to maximize their profits. The complex user interfaces, a language that is not user-friendly, and lack of tracking of payments also present a challenge to many farmers with regard to registration, transaction verification and tracking. The limitations are overcome in the proposed system which proposes a bidding-based mobile marketplace which targets small and medium-scale farmers, specifically. The platform is more inclusive, accessible, and transparent than the previous technologies with such features as

OTP-based signalization, multilingualism, and real-time bidding.

1.4 Proposed Approach

The mobile application suggested fulfills the purpose of a digital connection between farmers and verified buyers, which implies the transparency and equity of each transaction. The farmers are able to register easily with their mobile number and also make a profile and post information about their produce including the type of crop, quantity, starting price, and availability. Customers are then able to search in real-time and post a bid to compete with prices that are seen as real-time. Upon expiry of the bidding period, the system automatically selects the best bidder and a secure payment procedure is launched by the system with an inbuilt gateway. The platform is developed based on Android Studio to do front-end coding and Firebase to do back-end programming that includes the authentication, real-time database, and cloud storage. The system relies on Firebase Cloud Messaging (FCM) typing through which instant-updates are provided and buyers and farmers are kept updated about what is going on with bidding, payment and confirmations. This development is modular, and the reason is that it can be extended to accommodate scalability and easy maintenance. The system may be further developed in the future with more high-tech features, including AI-generated price prediction, predictive crops, and thematic interconnections to IoT-based sensors to monitor that processes take place in time. The primary objective of such strategy is to turn the platform into one that is dependable, flexible, and can serve agricultural communities on a massive scale.

1.5 Objectives

The main objectives of the project are:

1. Enable direct and transparent selling for farmers

Build an easy-to-use mobile application to remove the intermediaries, enable competitive pricing discovery through fair competition and enhance the farmer participation by ease-of-use

interfaces and providing languages.

2. Ensure secure and efficient transactions

Combine secure electronic payment systems and real-time bid, sales and confirmation of payment notifications to ensure the transaction process is smooth.

3. Build a scalable and reliable backend

Architecture A secure and scalable backend should be designed to be based on Firebase Firestore

and Cloud Functions to facilitate a smooth operation and eventual growth.

1.6 Sustainable Development Goals (SDGs)

The project is consistent with a number of the United Nations Sustainable Development Goals (SDGs), in particular, with those related to the reduction of poverty, innovation, and sustainable development:



Figure 1.1 Sustainable Development Goals

SDG 9: Industry, Innovation, and Infrastructure – The project supports digital innovation in agriculture by creating a technology-driven marketplace that enhances connectivity between farmers and buyers. It strengthens rural digital infrastructure and promotes the adoption of modern, efficient systems for market access.

SDG 12: Responsible Consumption and Production – The platform encourages sustainable agricultural practices by reducing middlemen-related wastage, enabling better demand forecasting, and promoting transparent, responsible buying and selling between farmers and consumers.

1.7 Overview of Project Report

The following report displays the entire development procedure and assessment of the planned Mobile App Direct Access to Farmers. Chapter 1 will contain the introduction, background and the objectives of the project and how it is aligned to the SDGs. Chapter 2 is a literature review of the available literature and technology that informed the design of the project. The system architecture, data flow diagrams, and module descriptions are

described in chapters 3, discussing the main functionalities of the application. The 4th chapter is dedicated to the implementation information, which addresses tools, frameworks, and algorithms employed in the course of its creation. Chapter 5 will include the results and discussion which includes. performance measurements of the system, and the more in-depth analysis of user interaction. Chapter 6 is the final part of the report where the most important findings and limitations are presented, as well as opportunity to improve the report.

The project report, therefore, gives an overall picture of how the system was conceptualized, designed, implemented, and assessed to respond to the real situation facing the farmers in trying to gain access to the fair and open markets.

CHAPTER-2

LITERATURE SURVEY

2.1 Mobile App for Direct Market Access for Farmers by

Aniket Chauhan, Kalpana K Harish, Utkarsh Kumar, Raunak:[1] This study proposes a mobile app that has been created to fill this gap between farmers and consumers by removing intermediaries, which in most cases diminishes the profits earned by the farmers. The application provides farmers with a convenient platform to post and update their farm products, predict prices based on AI-based algorithms, and optimize the delivery routes with the help of GPS integration. The application is aimed to make the process of using it as easy as possible, using clear interface, and navigation; the authors are sure of the precision and usefulness of AI-powered price prediction, which will help farmers take the right decisions in unstable markets. In parallel with it, the use of blockchain technology is used that can guarantee secure, trustful transactions, providing the buyers and sellers of the platform with even greater confidence. The challenges outlined can be summarized as the necessity to have high-quality and accurate data to use AI, as well as the challenges of scalability of blockchain technology in rural and connectivity-restricted areas. The research paper recommends that further enhancement in technology infrastructure and training of farmers remains the next important step towards the full exploitation of such digital interventions in agriculture.

2.2 AI-Powered Mobile Applications for Smallholder Farmers: Transforming Agricultural Productivity, Market Integration, and Resource Optimization

Vivek Kumar Soni, Roshan Verma, Sakshi Verma, Vasu Dewangan [2] The paper explores how mobile applications, which are driven by artificial intelligence, can be used to assist smallholder farmers. The authors highlight the significance of these applications in availing important agricultural data on time, including the weather, crop prices, pest and disease control and market connection. Through the use of mobile technologies, the study reports evidence of amplified productivity and financial inclusion with small-scale farmers, which ultimately improves rural livelihoods through the integration of different AI-based applications such as predictive analytics to forecast the demand and price, personalized recommendations, and optimization of resources. These are AI-powered assistants. farmers in determining data-driven choices, wastages, and better harvests.

The paper is discussing pilot studies that show that such applications can assist farmers to raise the level of income and enhance more effective integration into the market, although it is observed that the barriers to the effective use of smartphones and digital literacy through the lack of internet in remote locations are not completely eradicated yet. To ensure the greatest transformative effects of AI-driven mobile technologies in agriculture, the authors recommend long-term work on infrastructure enhancement and specific training programs.

2.3. Mobile Application to Direct Marketing Access for Farmers

Dr. Latha P H, Ms. Amulya R, Mr. Darshan GS, Ms. Aabha Pandey, Mr. Vinayaka C [3] The paper discusses a specialized mobile application that is intended to directly access farmers to consumers and markets and help them to eliminate the middlemen and increase profit margins. The application also has built in secure payment gateways and data synchronization features that allow smooth transactions and seeking of clear information in the market. It is customized to suit farmers with varying technical needs by its intuitive user interface.

Much focus is devoted to the fact that AI is integrated to give price and crop decisions, which assists a farmer in efficient marketing and production choices. It also uses weather APIs to assist users in the organization of farming endeavors according to weather conditions. All these functionalities help in enhancing efficiency in the market and the financial security among farmers.

Nevertheless, the authors identify the number of barriers, such as the connection problem in rural areas and insufficient levels of digital literacy among the users, as potential threats to mass adoption. Besides, the reliance on third-party payment platforms such as PayPal creates weaknesses that require considerate mitigation mechanisms. The paper postulates that future improvements in digital literacy and local infrastructure will play a key role in conquering these obstacles.

2.4. AI-Driven Platform for Direct Market Access for Farmers

Rupali Vyas, Vedita Kesharwani, Chirag Rohra, Rounak Sundrani [4] The presented paper proposes a more advanced AI-based online market that targets empowering farmers by granting them direct access to the consumers and farming enterprises. The platform makes use of machine learning algorithms to provide real-time price optimization, individual crop

suggestions, and forecasting the demand.

Some of the features that are incorporated in its design are the use of different languages and chatbots to make it accessible to different groups of users, addressing the needs of people with varying linguistic and digital capabilities. The study also mentions the fact that the platform enhances transparency and negotiation strength to farmers because they gain access to comprehensive market intelligence and automated inventory control. AI algorithms are an essential part of improving the user experience through the dynamic optimization of pricing strategies that depend on market trends thus maximizing farmer profits.

Although innovative in nature, the authors do not ignore issues like the fact that, powerful digital infrastructure is necessary to facilitate AI features and that large-scale training on digital literacy is also required to facilitate its successful utilization. Inclusivity and scalability are also a consideration to ensure the platform can affect the rural agricultural communities to a larger extent.

2.5. Blockchain Technology in Agriculture: A Case Study of Blockchain Start-Up Companies

Chandra Sekhar Bhusal [5] The paper by Bhusal discusses the use of blockchain in agriculture, with the author drawing upon start-up firms where the technology has been applied to tackle the problem of supply chain fraud, transparency and traceability using decentralized ledgers and smart contracts. The research highlights that blockchain has a potential to generate tamper-free records that can instill trust in the relationship between farmers, intermediaries and consumers to enhance effectiveness and security of transactions.

The paper records a number of case studies that demonstrate the use of blockchain in eliminating price fluctuations, counterfeit agri-products, and in securing payments in a timely manner by use of automated contractual agreements. These inventions are the way forward towards a more resilient and open agricultural economy particularly because smallholder farmers are at a level playing field.

As much as the potential gains are high, Bhusal also addresses the challenges facing the use of blockchain with such challenges including high cost of implementation, issues of scalability and limited knowledge of many agricultural players with regard to technical aspects. The paper suggests that these limitations need to be overcome by efforts in collaborations; policy support and education of the stakeholders.

2.6. Emerging Opportunities for Blockchain in the Agri-food Industry

Giuseppe Tripoli, Jürgen Schmidhuber [6] In this paper, the author discusses the disruptive nature of blockchain technology in the agricultural and food industries. The authors explain how the essential features of blockchain such as decentralization, transparency, and immutability may be used to overcome some of the significant issues such as fraud, absence of traceability, and ineffective transactions. The paper provides an overview of various applications of blockchain to supply chain integrity that enables consumers and producers to trace the originality and authenticity of agricultural products.

The paper is substantially dedicated to smart contracts and the Decentralized Finance (DeFi) systems that present new opportunities of inclusive finance within rural agricultural communities. The direct peer-to-peer transaction, lower operational expenses, and the availability of transparent financial records can be enabled by blockchain-based platforms, and especially farmers not provided with access to the usual banking services would find it useful. Although optimistic on the advantages of blockchain, Tripoli and Schmidhuber note that challenges like technological complexity, uncertainty in legislation and scale challenges are some of the challenges that need to be overcome to allow wider adoption. They also promote a partnership between policymakers, technology creators, and agricultural stakeholders to create an enabling agriculture blockchain environment.

2.7. Transforming Agricultural Supply Chains: Leveraging Blockchain-enabled Java Smart Contracts and IoT Integration

Adil El Manea, Khalid Tatane, Younes Chihab [7] This study introduces a compounded model of using blockchain smart contracts and Internet of Things (IoT) technology to transform agricultural supply chains. The suggested system automatizes the settlements of the transactions between farmers, traders, and consumers. IoT gadgets gather farm and produce information in real-time, which is verified through blockchain networks to achieve data fidelity and improve the visibility of the supply chain. The authors show that the method minimizes human error, turns the transaction confirmations faster, and the risk of fraud is decreased. The platform provides timely and trustless settlements by using the power of Java-based smart contracts, which impose predetermined conditions and to automatize payments. This smooth combination of the IoT and blockchain is presented as fundamental to the current state of accuracy in agriculture and effective supply chains management.

Among the limitations, one can distinguish the complexity of the system and the necessity of

large-scale implementation of the IoT infrastructure, the limitations to which must be supported by effective technical and policy frameworks. The paper ends by supporting the fusion of this technology as one of the key innovations to sustainable and responsible agricultural trade..

2.8. Mobile Application to Direct Marketing Access for Farmers

Dr. Latha P H, Ms. Amulya R, Mr. Darshan GS, Ms. Aabha Pandey, Mr. Vinayaka C [8]

The paper discusses a mobile application that is set to ensure that the farmers have direct access to the marketing platform so that they are not driven by the middlemen. The application combines real-time prices of crops, payment security, and efficient inventory control, which facilitate smooth business between farmers and consumers. The unique advantage of the application is that it is multilingual and oriented towards user-friendly interfaces and addressing the needs of farmers of different technological abilities. The paper highlights the significance of digital payments through integration with such services as PayPal that enhance the security of the transaction and its expediency. It is also built with weather and AI-based crop recommendations, which enhances the decision-making of farmers and mitigates risk. The authors express the issues in the areas where the internet connection is low and that farmers should be trained to take advantage of the full potential of these mobile digital solutions.

In conclusion, the paper recommends that although mobile applications present great opportunities to revolutionize agricultural marketing, the issue of infrastructures and digital literacy still poses a critical challenge.

2.9 Summary of Literature Reviewed

Table 2.1 presents a comprehensive summary of the key findings, methodologies, and Limitations identified in the reviewed literature.

Reference	Focus Area	Key Findings	Limitations
Aniket Chauhan, Kalpana K Harish, Utkarsh Kumar, Raunak [1]	Mobile app for direct market access	Android app enables farmers to list crops, connect directly with buyers, and manage orders and payments, reducing intermediaries and improving transparency.	Requires smartphones and stable internet; adoption depends on farmers' digital literacy.

Dr. Latha P H, Amulya R, Darshan G S, Aabha Pandey, Vinayaka C [2]	Direct marketing mobile app with payments	App integrates crop listing, buyer search, and secure digital payments to support direct marketing with a simple UI for farmers.	Relies on third-party payment gateways; connectivity issues in rural areas can disrupt use.
Rupali Vyas, Vedita Kesharwani, Chirag Rohra, Rounak Sundrani [3]	AI-driven direct market platform	AI platform optimizes prices, offers crop recommendations, and provides multilingual UI and chatbots to improve farmer-buyer interaction.	Depends on continuous data and robust backend; rural connectivity and training remain hurdles.
Chandra Sekhar Bhusal [4]	Blockchain in agriculture (case studies)	Case studies show blockchain improves traceability, prevents fraud, and secures agri-transactions via distributed ledgers and smart contracts.	High deployment cost; low blockchain awareness among farmers; regulatory and scalability challenges.
Giuseppe Tripoli, Jürgen Schmidhuber [5]	Blockchain in agri-food systems	Reviews how blockchain can enhance transparency, traceability, and inclusive finance in agri-food value chains.	Evidence mostly from early pilots; issues of scalability, interoperability, and regulation remain.
Vivek Kumar Soni, Roshan Verma, Sakshi Verma, Vasu Dewangan [6]	AI-powered mobile apps for smallholders	Describes AI in mobile apps for productivity, market integration, and resource optimization through recommendations and decision support.	Impact constrained by smartphone cost, network availability, and digital literacy gaps.
Adil El Manea, Khalid Tatane, Younes Chihab [7]	Blockchain + IoT supply-chain automation	Integrates Java smart contracts with IoT sensors to automate settlements and validate agricultural supply-chain data end-to-end.	Complex architecture; requires IoT infrastructure and technical expertise often missing in villages.
S. V. S. Ramakrishnam Raju, Bhasker Dappuri, P. Ravi Kiran Varma, Murali Yachamaneni, D. Marlene Grace Verghese, Manoj Kumar Mishra [8]	Smart farming Android app (KisanVishwa)	Presents an intuitive Android app linked to IoT/AI modules for monitoring, crop advice, and smart farming workflows.	Evaluated on limited users; long-term adoption and performance at scale untested

M. W. Gouroubera et al. [10]	Mobile-phone-mediated agri-technologies	Analyzes drivers and impacts of mobile-phone-based scaling of agricultural technologies on yield, profit, and learning outcomes.	Benefits vary by context; constrained by network coverage, service design, and farmers' ability to act on advice
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Table 2.1 Summary of Literature Reviews

2.10. Identified Gaps and Research Opportunities

Digital Literacy Challenges: Most farmers in rural India still struggle with using digital applications because they are not familiar with smartphones. Existing interfaces are often text-heavy and not designed for low-literacy users. This creates a barrier, preventing farmers from benefiting from digital market platforms. Applications also lack intuitive workflows, making it difficult for first-time users to navigate. As a result, adoption remains low despite the availability of digital tools.

Dependency on Middlemen: In most agricultural markets, middlemen dominate price negotiation, leaving farmers with limited bargaining power. Digital platforms rarely provide transparent, real-time pricing comparisons for crops. This allows intermediaries to manipulate prices, reducing earnings for farmers. Without visibility into market rates, farmers cannot take timely decisions. The system remains unfair and inefficient.

Inadequate Bidding or Auction Systems: Only a few platforms support open and competitive bidding mechanisms. Even those that exist lack clarity in how bids are ranked or how winners are selected. The absence of automated bid verification leads to doubt and disputes. Farmers need a transparent system where bids are time-stamped and fairly visible. This is an important technological gap.

Voice-Based and Multilingual Interfaces: Voice assistants in regional languages can drastically improve accessibility. Speech-based commands simplify usage for low-literacy farmers. Offline-capable voice recognition can help in areas with poor internet connectivity. Providing real-time translation of crop details can support farmer–buyer communication. This area has strong potential for innovation.

CHAPTER-3

METHODOLOGY

3.1 Overview

The research design used in this project is systematic and modular in order to ascertain that the mobile application is produced efficiently, its tests are conducted comprehensively and its implementation is efficient. The subject of this approach is the development of a real time, secure and easy to use communication solution between farmers and the buyer. These projects are based on the Software Development Life Cycle (SDLC) model and puts an emphasis on iterative development. All the phases, i.e., requirement gathering, design, implementation, and testing, were carried out in a systematic order with feedback incorporated after each phase. This was how the application would stay in line with the expectations of the users and be flexible to make improvements in the future.

3.2 Methodology Adopted

The system proposed has been built through the use of Agile methodology that enables incremental development and will constantly receive feedbacks by the users. This model was selected since it supports the changes of a design to be flexible; it promotes teamwork and ensures that the effective components are delivered in a rapid manner. Every sprint was made up of planning, development, testing and evaluation. Requirement Analysis - Determining both the functional and non-functional requirements based on the user interviews, literature review, and problem analysis. System Design -Bringing the functional flow to life through the Android Studio and integrating Firebase to perform the authentication, database and cloud storage functionality. Testing -Testing of the system by conducting both unit, integration and system testing to ensure that the system met the project objectives. Deployment -Deploying the application to a sandbox environment and connecting it to firebase to perform the authentication, database and cloud storage functionality. Evaluation -Gathering user feedback and performance results to verify.

3.3 Tools and Technologies Used

The project integrates both front-end and back-end technologies to create a smooth and responsive application.

Front-end: Android Studio (Java), XML-based UI design

Back-end: Firebase Firestore (database), Firebase Authentication, Cloud Functions, Firebase Storage

Notifications: Firebase Cloud Messaging (FCM)

Payment Integration: Secure API-based payment gateway (sandbox mode) **Version Control:** Git and GitHub

Design Tools: Canva and Figma for UI prototyping and mockup design

The combination of these technologies ensures that the application is scalable, reliable, and efficient in real-time operations.

System Modules

The system has been structured into a number of clearly defined modules that are in charge of processing a certain aspect of the functionality within the application. It is a modular structure that allows the development process to be well-organized and that the future updates or additions can be done without disrupting the whole system. All modules work in harmony with one another to provide a one-stop experience to both the farmers and buyers.

Authentication Module:

This module has the task of establishing the identity of the users to gain entry into the system. Authentication is done via the OTP-based login method used by farmers and email or phone-based login used by buyers. The module is constructed using Firebase Authentication and it makes sure that only the legitimate and verified users are involved in the platform. It also takes care of user role and redirects them to the right dashboard depending on them being farmers, buyers or administrators.

Farmer Module:

The farmer module enables registered farmers manage their agricultural listing as well as monitor bidding. Farmers are able to add new crops by filling the information like crop name, quantity available, base price and location with corresponding images uploaded. They have an opportunity to see the bids made in real time on their produce and complete transacting when an acceptable offer is made. The module has been built in a straightforward and understandable manner so that farmers who have little or no digital exposure can be able to utilize the application.

Buyer Module:

The buyer module assists buyers in the process of browsing, and buying of agricultural products by farmers. Customers are able to see all current crop prices, compare and bid on crops of interest. The interface is also designed with the intention of displaying key data points that include the type of crop, quantity, minimum price, and the location of the seller. When a buyer makes a bid the module notifies them about bid status change and auction result via real time notifications.

Bidding and Auction Module:

This is the main module of this application and handles all the operations related to bidding. It approves all the incoming bids and cross-checks them against the existing highest bid to ensure fairness. Firebase Firestore and Cloud Functions will allow updating bids in real time and closing the auction automatically, depending on predetermined time restrictions. This module will make sure that the synchronized and updated information is visible to all the users, and the bidding environment is competitive and transparent.

Payment Module:

When an auction has been completed and the farmer has taken the final price, the payment module makes the buyer and the farmer transact through it. It incorporates a secure payment gateway which receives payment under a sandbox testing environment. Having made the payment, the buyer is checked by the module that everything went right, and the transaction is registered in the database. It guarantees safe money transfer and minimizes the risk of having cash that is manually being handled to process payments.

Notification Module:

All real-time communication between the system and the users is handled by this module. It is created with Firebase Cloud messaging, which sends notifications about events like the placing of bids, auction ends, and payment receipts and image or text updates of the farmer or buyer. This module can increase the engagement of the user by making sure timely notifications are made to both parties and therefore keep them updated during the cycle of the transaction.

Admin Module:

The admin module gives management and control of the activities in the platform. The

administrators are able to check the user accounts, track the active listing, check the bidding activity, and interfere in case there is suspicious or unauthorized actions. This module guarantees that the site is safe, reliable, and it is not violated by operational policies.

Combined, these modules form a digital market that is coherent and makes agricultural trading an easy task, facilitates transparent bidding and allows secure transactions. The modular design does not only enhance efficiency in the development process but also makes the system scalable and open to further developments in the future like AI-powered price prediction or even tracking the logistics.

3.4 Development Process

The process of development started with the identification of the user personas: farmers, buyers, and administrators and defining the interactions between them. These user flows led to the development of separate modules of registration, crop listing, bidding, notifications, and payment, with the application being divided according to the separation of logic, UX, and data layers to make it more maintainable. The choice of Firebase Firestore as the backend is explained by its capacity to provide real-time synchronization and offline data retention that can be very important to users in connectionally challenging locations. The features were implemented and tested step by step to make their functionality smooth. The constant integration was with GitHub, which facilitated version control and other developers could also work together in developing the project and tracking changes easily during the project lifecycle.

3.5 System Architecture

The system architecture of the Mobile App is designed to provide a reliable, scalable, and user-friendly digital platform that connects farmers directly with buyers, eliminating the need for intermediaries. The architecture follows a three-tier structure comprising the presentation layer (frontend mobile interface), application layer (Firebase services and logic control), and data layer (cloud-based storage and database). This design ensures modularity, real-time synchronization, and secure communication between users and backend services. At the presentation layer, the system uses an Android-based mobile application developed in Java within the Android Studio environment. This serves as the user interface through which farmers, buyers, and administrators interact with the system. Farmers can create listings, upload crop details, set base prices, and monitor bids in real time, while buyers can browse listings, place bids, and complete payments. The interface is designed with simplicity and accessibility in mind, ensuring usability even for

users with minimal technical literacy. Multilingual support and large icons enhance the overall user experience, particularly for rural users. The application layer acts as the bridge between the frontend and the database, handling business logic, user authentication, and automated event management. This layer is powered by Google Firebase, which provides core backend services such as Authentication, Firestore Database, Cloud Functions, Cloud Storage, and Cloud Messaging.

Firebase Authentication manages secure user login through email/password access for buyers and formers.

Firebase Cloud Firestore stores all dynamic data, including user profiles, crop listings, bids, and transactions, ensuring real-time data synchronization across devices.

Firebase Cloud Functions execute automated backend operations, such as validating bids, determining auction winners, and updating payment statuses. Firebase Cloud Messaging (FCM) handles push notifications, keeping users updated on bidding activity, auction results, and payment confirmations. Firebase Cloud Storage stores multimedia assets such as crop images and profile pictures securely, providing easy retrieval through URL mapping.

The data layer comprises the structured storage and retrieval mechanisms that ensure data integrity and persistence. Firestore serves as the core database, organizing data into collections and documents for efficient query execution. Acasts are indexed and replicated by default in the distributed cloud structure of Firebase to give it scalability and fault tolerance. This makes certain that despite any latency occurred in one node of the system, all users are responsive and consistent. In addition to Firebase, the architecture integrates a secure payment gateway API to handle monetary transactions between buyers and farmers. The payment system operates in a sandbox mode for testing, simulating real-world payment scenarios without processing live financial data. When a buyer completes a payment, the gateway securely processes the transaction and communicates the result back to Firebase via a webhook. A corresponding Cloud Function updates the payment status in the database, ensuring transactional accuracy and consistency. The communication between layers takes place over HTTPS (TLS protocol), guaranteeing secure data transmission and preventing unauthorized interception. The app also uses Firebase Security Rules to define strict access permissions, ensuring that only authenticated and authorized users can view or modify specific data. The architecture supports scalability and offline functionality. Firestore's offline persistence allows data to be temporarily stored on the device when connectivity is poor, with automatic synchronization once the internet connection is restored. This feature ensures uninterrupted usability, which is particularly important for users in rural areas with limited network access.

Figure 3.1 illustrates the overall system architecture of the mobile application. It shows how each

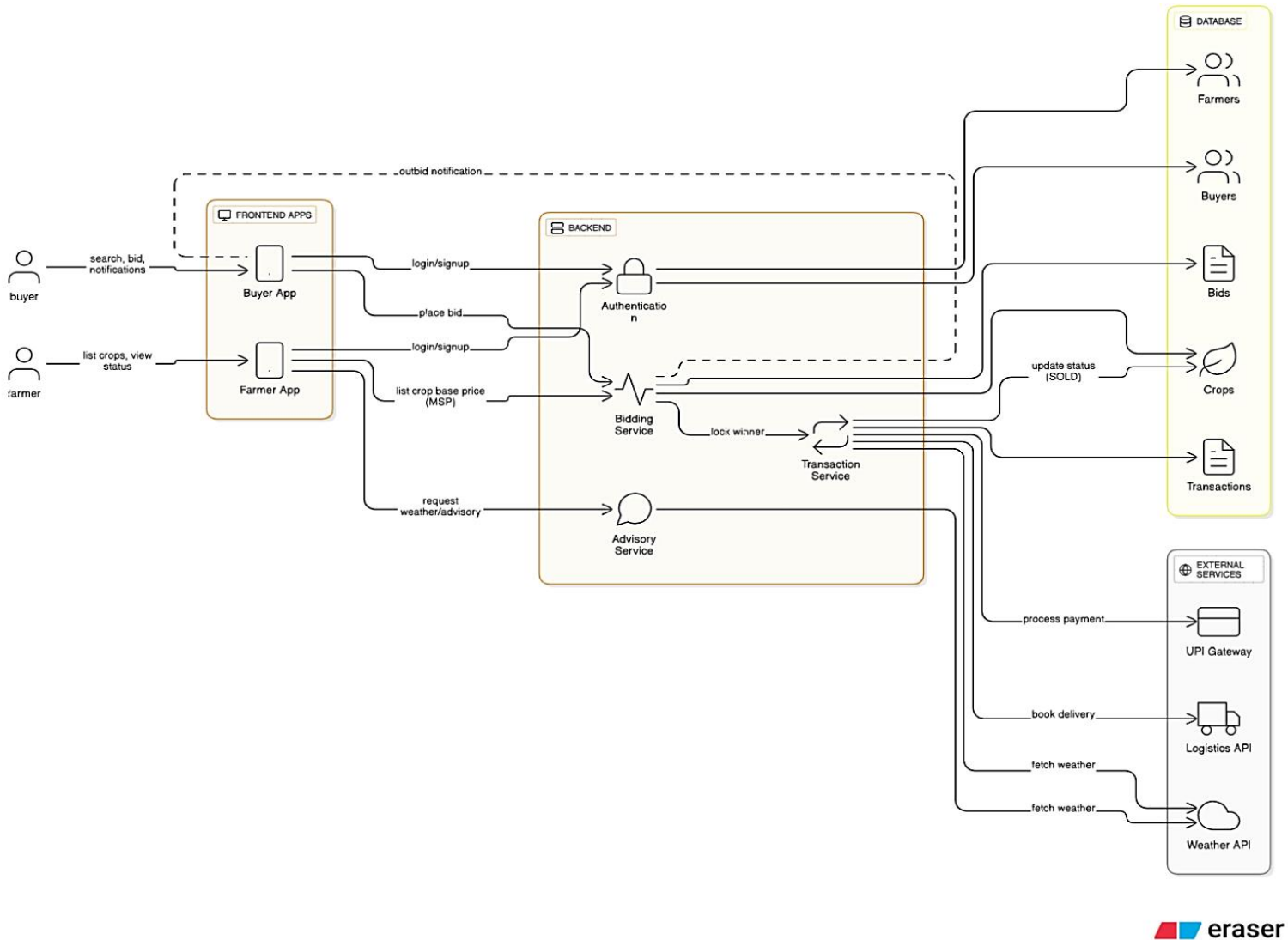


Figure 3.1: System Architecture of Mobile App for Direct Access for Farmers

Overall, the architecture ensures high performance, data consistency, and a secure transaction environment. It provides a modular structure that allows future enhancements such as AI- driven price prediction, blockchain-based traceability, or integration with government digital marketplaces. This layered and cloud-centric design establishes a strong technical foundation for scalable, sustainable agricultural digitalization.

CHAPTER-4

PROJECT MANAGEMENT

Project management is a critical component of any successful software development effort. It provides the structure required to plan, organize, and control project activities to ensure that goals are achieved within the defined scope, time, and budget. The primary aim of project management in this context was to maintain a clear workflow from requirement analysis to deployment. Since the project involved both technical and user-centric objectives, effective management ensured that each stage of development aligned with the overall vision of empowering farmers through digital technology. The Agile methodology was adopted for managing this project because of its flexibility and adaptability. Unlike rigid waterfall models that rely on fixed sequences, Agile allows continuous feedback and incremental improvement. Each iteration, or “sprint,” resulted in a partially functional module, which was reviewed and refined before moving to the next stage. This not only reduced the chances of large-scale errors but also made it easier to respond to changing requirements or feedback from the guide and end users. The project team maintained weekly discussions and review meetings with the guide to track progress and address any challenges encountered during development. This approach ensured that the work was divided equally among members, deadlines were met, and quality was maintained. Each milestone— requirement analysis, UI/UX design, backend setup, bidding logic implementation, and payment integration—was completed as part of a planned sprint, leading to the successful development of the final application.

4.1 Project Timeline

The success of a software project depends heavily on proper scheduling and time management. The project timeline was carefully structured to balance academic deadlines with realistic development goals. The entire duration of the project extended over five months, divided into

specific phases: planning, system design, development, testing, and evaluation.

The planning phase focused on understanding the existing agricultural market ecosystem and identifying the key problems faced by farmers. This phase involved extensive literature review, interaction with subject experts, and studying related mobile applications. The goal was to clearly define the system's requirements and establish a foundation for design.

In the design phase, the system's architecture, user interface, and database schema were developed. Tools such as Figma and Canva were used to visualize app layouts, ensuring that the design remained simple and intuitive for users with limited technical experience. The Firebase database structure was modeled to accommodate multiple user roles (farmer, buyer, and admin) and maintain real-time synchronization.

The implementation phase involved developing the mobile app in Android Studio and integrating it with Firebase for authentication, data storage, and cloud operations. This stage was carried out in multiple iterations to ensure the smooth functioning of each module. The bidding module, for instance, underwent several cycles of testing and refinement to ensure accurate real-time updates and proper handling of auction data.

In summary, the timeline was designed to balance theoretical research with practical implementation. The breakdown of the project schedule is shown below (approximated textually):

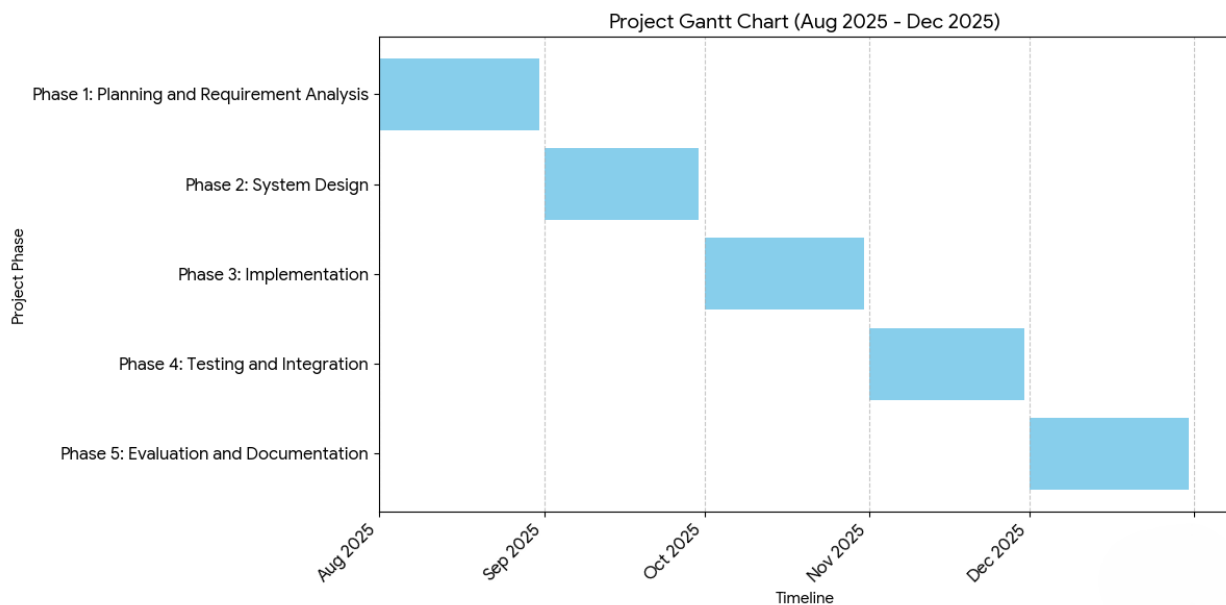


Figure 4.1: Project Time Line

The Gantt chart corresponding to this timeline highlights the interdependence of each activity, ensuring that milestones were met within the planned timeframe. Regular progress reviews helped maintain pace and allowed for adjustments when certain stages required additional time for debugging or performance optimization.

4.2 Team Roles and Responsibilities

Darshan B — Team Lead & Backend Integration

Darshan acted as the team lead and coordinated the overall progress of the project. He handled Firebase setup, including Authentication and Firestore database configuration. He worked on connecting the backend with the application, especially for features like crop listing, bidding, and user login. Darshan also prepared the system architecture diagrams and helped integrate the payment gateway in test mode. In addition, he supported the team during debugging and ensured that the project stayed on track with the deadlines.

Sujith M Maddin — Front-End & App Interface Development

Sujith mainly worked on the front-end development of the mobile application. He designed and implemented the user interfaces using XML in Android Studio and worked on screens such as login, dashboard, crop listing, and bidding. He focused on making the app easy to use and visually clear. Sujith also worked closely with the backend integration by connecting UI components to Firebase data and testing the overall flow of the mobile application.

Nagashree M N — Documentation & Testing

Nagashree was responsible for preparing the project documentation, including the report, review presentations, and formatting work. She ensured that all sections were written clearly and according to the required structure. Nagashree also tested the application throughout the development stages to make sure the features worked properly. She verified login flow, listing updates, bidding functions and noted issues for the team to fix. Her contribution helped improve the overall quality and correctness of the system.

Collaborative Work

All team members participated in discussions, literature review, presentation preparation, and testing. Important decisions about design, data flow, and implementation were taken together. Each member supported others whenever needed, which made the development process smoother and more manageable.

4.3 Risk Analysis

Every software project involves certain uncertainties and potential obstacles that can affect progress or outcomes. Identifying and managing these risks early is essential to ensure smooth execution. In this project, the primary risks were categorized as technical, operational, schedule-related, and user-related.

The first major risk was technical dependency. Since the system heavily relied on Firebase for authentication, real-time database updates, and notifications, any downtime or API limitation could disrupt functionality. To mitigate this, offline caching was implemented to allow data access even when the device temporarily lost connectivity. Regular synchronization ensured data integrity once the network was restored.

The second risk involved data security and privacy. As the application handled sensitive user information such as phone numbers and payment records, proper security measures were enforced. Firebase's in-built authentication and encryption mechanisms were configured to prevent unauthorized access. Firestore security rules were carefully written to ensure that users could only access their own data, minimizing the risk of data leaks.

Another critical challenge was time management. Since the project was developed alongside academic coursework, meeting deadlines required disciplined scheduling. Delays in one phase could impact subsequent tasks. Frequent testing during development also prevented accumulation of unresolved bugs toward the end of the project.

To address this, the application was designed with minimal text, large icons, and clear navigation. Local language support was also considered to improve accessibility. Simple onboarding tutorials were included to help first-time users understand how to list crops or place bids.

Lastly, payment integration and testing carried operational risks. Since real transactions were not performed in the prototype phase, a sandbox environment was used to simulate payments. This prevented security breaches while allowing functional verification of transaction workflows.

Through proactive risk management and continuous communication, the team was able to overcome these challenges without compromising project quality or delivery timelines. The Agile model also made it easier to identify risks early and implement corrective actions promptly.

4.4 Summary

Project management ensured that the development of the Mobile App for Direct Access for Farmers was structured, efficient, and outcome-oriented. Through careful scheduling, realistic budgeting, and proactive risk handling, the project was successfully completed within the stipulated time. The Agile approach allowed flexibility and iterative improvement, while risk analysis and budget management helped maintain stability throughout the project life cycle.

This disciplined approach ensured that the project not only met its technical objectives but also maintained alignment with its broader goal — empowering farmers through digital innovation and creating an accessible, sustainable agricultural trading ecosystem.

CHAPTER-5

ANALYSIS AND DESIGN

System analysis and design form the foundation of every successful software project. This stage involves understanding what the system must achieve, translating user needs into clear requirements, and converting these into a structured system architecture and workflow. The Mobile App for Direct Access for Farmers was designed to be both technologically robust and user-friendly, offering a seamless platform for direct trade between farmers and buyers. The analysis and design process aimed to ensure that every functional component of the system operates in harmony, delivering a reliable and efficient experience for end-users.

The goal of this phase was to thoroughly analyze the requirements, identify the components necessary for implementation, and design the logical and physical architecture of the system. Proper analysis helps reduce errors during development, ensures scalability, and maintains performance under varying loads.

5.1 Functional Requirements

The functional requirements define the core operations and behaviors that the proposed system must support. These requirements are based on problem analysis, literature review, user feedback, and practical agricultural challenges.

Farmer Registration & Profile Management

- The system must allow farmers to **register and log in** securely using their email, phone number, and password.
- Farmers should be able to **create and update their profiles**, including name, contact details, address, and location.
- The application should enable farmers to **manage their accounts**, update personal information, and recover their passwords if required.

Crop Listing and Management

- Farmers must be able to **add new crop listings** by entering details such as crop name, quantity, base price per kilogram, and uploading images.

- They should be allowed to **edit or delete existing listings** at any time.
- The system must display the **current status** of each crop (Open, Closed, Sold).
- Farmers should be able to **track bidding activity** and monitor buyer responses in real time.

Buyer Browsing and Search Functionality

- Buyers should be able to **view all available crops** on the platform.
- The system must support **searching and filtering** by crop name, location, and price.
- Buyers should have access to **detailed crop information**, including images, base price, quantity, and farmer details.

. Bidding and Negotiation Module

- Buyers must be able to **place bids** that meet or exceed the farmer's base price.
- The platform must provide **real-time updates** on bidding activities.
- Buyers should receive **notifications** when they are outbid.
- Farmers must have the ability to **review all bids**, confirm the highest bidder, and close the bidding process.

5.2 Non-Functional Requirements

Non-functional requirements define how the system should perform rather than what it should do. The project emphasizes reliability, scalability, and user accessibility to ensure long-term sustainability.

The reliability requirement ensures continuous system availability, supported by Firebase's cloud infrastructure and data synchronization features. From a usability perspective, the interface is designed to accommodate farmers with minimal technical experience. This includes clear iconography, larger buttons, and minimal text input. The security requirement is fulfilled through Firebase Authentication and Firestore's access control rules, which ensure that only authorized users can view or modify specific data. Scalability is another crucial aspect—Firebase's cloud-based nature allows the system to handle an increasing number of users and transactions without major architectural changes. The system's maintainability is ensured through modular coding practices, enabling developers to add or modify features without affecting the existing functionality.

5.3 Software Requirements

The project utilized a combination of cloud-based and local development tools to achieve high performance and efficiency. The software requirements include:

Android Studio: Used as the primary Integrated Development Environment (IDE) for building and testing the mobile application.

Firebase Services: Employed for authentication, real-time database management, storage, and cloud functions.

Java: Programming language used for building the app logic.

Canva/Figma: Used for designing the user interface mockups and prototypes.

GitHub: Utilized for version control and collaborative development.

Postman: Used to test and verify backend API calls and cloud functions.

These tools were chosen for their compatibility, scalability, and ease of integration, which helped streamline the development and deployment process.

5.4 Hardware Requirements

The hardware requirements for the project were kept minimal since the application runs on Android devices and uses cloud storage for backend operations.

The development environment required a personal computer with at least 8 GB RAM, a quad-core processor, and 256 GB storage to support Android Studio's resource demands. For testing purposes, Android smartphones with versions 8.0 (Oreo) and above were used to ensure compatibility across devices.

The Firebase infrastructure managed all backend processing and data hosting, eliminating the need for physical servers. Internet connectivity was essential for synchronization, but core functionalities such as viewing cached listings and submitting bids were supported offline to enhance usability in rural areas with poor connectivity.

5.5 Human Requirements

The human resource aspect of the project involved three team members, each responsible for different domains: frontend development, backend integration, and testing/documentation. Collaboration and task tracking were maintained through GitHub and Google Sheets. The supervision of the project guide ensured proper alignment with

academic and technical standards.

Since the app targets rural users, a small-scale user study was conducted informally to understand farmer behavior and preferences. The insights gained helped refine features such as the bidding interface and notification structure to make them intuitive and accessible.

Summary

The requirements phase ensured that both user needs and system capabilities were clearly defined. By combining user-focused design, scalable cloud infrastructure, and modular coding principles, the system was built to deliver performance, security, and reliability. The identified requirements formed the foundation for the subsequent design and development phases.

5.6 Firebase Firestore Database Structure

The mobile application uses Google Firebase Cloud Firestore as its primary database. Firestore follows a NoSQL, document-collection data model, allowing flexible, real-time storage of structured records. For this project, three major collections were created: users, crops, and bids. Each collection stores multiple documents, and every document contains a set of key-value pairs that represent an individual entity such as a user, a crop listing, or a bid.

A. Users Collection

This collection stores the details of all users who interact with the system, including farmers and buyers. Each user entry is stored as a separate document with an auto-generated unique ID from Firebase Authentication. The key-value pairs in each document represent user attributes.

(default)	users	UYSGXJNFRhbYZ7q7XVMeZ5khUdW2
+ Start collection	+ Add document	+ Start collection
Crops	43W0Qs6VgDhznALQC7IjH0wAiWh2	+ Add field
bids	JAC1SXWI4pXSe0FhLgqc0xQ8sAJ3	address: "Mysore"
crops	KUZs2H1DN5djP34VD02c66dvHIV2	email: "dar@gmail.com"
users	07DPH0wqessnNTSsa3L8	name: "Darshan"
	QZ7fskDm2HTu8Q0I9ZxZ2LNhnJu2	role: "farmer"
	UYSGXJNFRhbYZ7q7XVMeZ5khUdW2	
	aXjj92jneRYF62ne2q7GnrhDUx22	
	bqPlfAwt4IaLFFeoxh9oZ7HLVjz2	
	maCf1yWxDCZbUnQAτXL4	
	nfgyTFbct2hpVmVHB0mo6go7Z7L1	

Figure 5.1 Example Documents in the users Collection

B. Crops Collection

The crops collection stores all produce listed by farmers. Each document represents a single crop listing and contains detailed attributes such as pricing, quantity, and availability. These details help buyers browse and filter listings easily.

(default)	Crops	71KKFlmORZI3uS6YKcoe
+ Start collection	+ Add document	+ Start collection
Crops >	1LtAoPH5YkvwszKrcRn	+ Add field
bids	5vL1o3qDudI3tMq1st7d	biddingEndDate: December 3, 2025 at 12:00:00 AM UTC+5:30
crops	71KKFlmORZI3uS6YKcoe >	createdAt: December 3, 2025 at 4:31:39 PM UTC+5:30
users	JXZjfVtdysNvofMtvv2b	farmerId: "QZ7fskDm2HTu8QOI9ZxZ2LNhnJu2"
	LE05KV2EbV8otW4gGbd0	farmerName: "Darshan B"
	P5F0QRSIBG4bWclSqZvo	farmerPhone: "8431589621"
	S5DmbFR2bA8DMr9e58Hu	harvestDate: December 3, 2025 at 12:00:00 AM UTC+5:30
	ejz6WHblnNmfyZzMhZAH	highestBid: 0
	fTZs0LCCqtBSkehRU31e	id: "71KKFlmORZI3uS6YKcoe"
	lIsdTHEF2w4h7BIUF5tM	imageUrl: ""
	o7iJ1ztIF1l63SY7Qvft	isSold: true
		location: "Chikkamagaluru"
		name: "Tomato"

Figure 5.2 Example Documents in the crops Collection

C. Bids Collection

The bidding module is implemented using the bids collection. Each document stores a single bid made by a buyer on a particular crop. These bids are monitored in real-time so that farmers can track the highest bidder and close the auction when required.

(default)	bids	D4ES7Xw7SEoAMiUEsyY8
+ Start collection	+ Add document	+ Start collection
Crops	A2L1W6xkT95KFCwQcWMy	+ Add field
bids >	CPR8KfsdBvoHYsR2PAwa	bidValue: 45
crops	D1oRtzwLqSvccM0SA80Y	cropId: "he2upAZDyKVeYHO0vqZj"
users	D4ES7Xw7SEoAMiUEsyY8 >	cropName: "Apple"
	HRLE0SXxh8X8SZYOLWFh	customerId: "43WOQs6VgDhznALQC7ljH0wAiWh2"
	VXh80LrRhUIGZnQ6ROFK	timestamp: September 28, 2025 at 8:02:33 PM UTC+5:30
	ZxVNIu7kBRmePqjn6Iva	
	d5XLJ1hDcCRD8JHs8oT8	
	eP9XVaUXGqXpPsXdugcS	

Figure 5.3 Example Documents in the bids Collection

5.7 Block Diagram

The block diagram serves as the foundational blueprint of the Mobile App for Direct

Access for Farmers. It provides a high-level visual representation of how the various system components interact with each other to achieve the overall functionality of the application. The primary purpose of this diagram is to simplify the understanding of the system's internal processes by dividing it into smaller, well-defined units that communicate through logical data flow and process exchange.

At the core of the system architecture lies the user interface, which acts as the main interaction layer between the user and the application. The app's interface is divided into two distinct modules—one for farmers and the other for buyers. Each module is designed to cater to its specific user type while maintaining uniformity in layout, color scheme, and navigation. Farmers use their module to register, upload details about their produce, set base prices, and monitor bids. Buyers can browse active listings, place bids, and make secure payments.

The application layer forms the backbone of the system, handling the execution of business logic and decision-making processes. This layer processes user requests, such as adding a crop, placing a bid, or checking notifications, and ensures that they are properly validated before being forwarded to the backend services. Functions such as bid management, transaction handling, and notification triggering are managed in this layer using Firebase Cloud Functions and Android services.

The database and cloud layer is powered by Firebase Firestore, which handles real-time data storage and retrieval. Each user interaction—whether it is a crop listing, a bid submission, or a payment confirmation—is immediately synchronized to the database. This ensures that all users see up-to-date information at any given time. Firebase Authentication secures user access, while Firebase Cloud Messaging (FCM) facilitates instant notifications about bids, payment updates, or auction closures.

The payment gateway module plays a crucial role in ensuring secure and seamless financial transactions. Once a buyer wins an auction, the payment process is initiated through an integrated payment API, which allows secure transfers between the buyer and the farmer. The gateway supports multiple payment methods, including UPI and digital wallets, and operates in sandbox mode during development to simulate real payment flows without actual fund transfers.

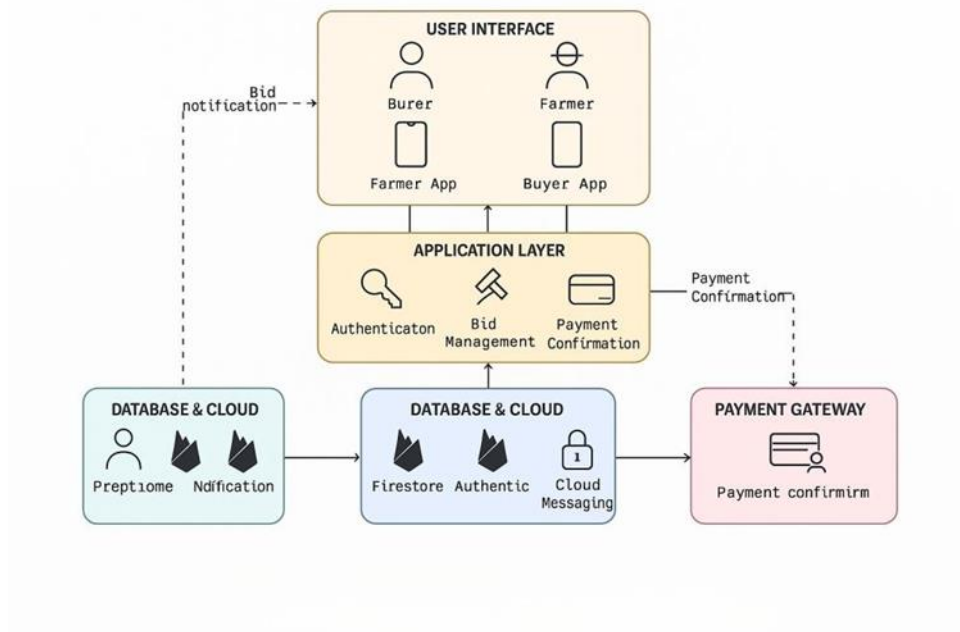


Figure 5.4: Application block diagram

Supporting all these layers is the admin and analytics module, which provides monitoring and regulatory control over system activities. The administrator verifies user authenticity, oversees system usage patterns, and manages reported issues. Additionally, the analytics component generates insights such as most-traded crops, average bid success rates, and farmer participation levels. This helps in evaluating the overall performance of the platform and planning future improvements.

The data flow within the block diagram follows a structured path. When a farmer lists a crop, the details are sent through the app interface to the Firebase backend, where they are stored in a structured collection. Once a bid is accepted, the payment process is initiated, and the system records the transaction while sending confirmations to both parties.

By organizing the system into interconnected blocks, the design achieves modularity and clarity. Each block can function independently, but together, they form a cohesive digital marketplace. The modular structure also simplifies debugging and maintenance since updates or changes can be made to one block without disrupting the others. Moreover, using Firebase's cloud ecosystem ensures that the system can scale automatically as user demand increases.

In summary, the block diagram effectively represents the complete workflow of the Mobile App for Direct Access for Farmers. It demonstrates how the user interface, backend services, and database interact seamlessly to create a transparent and reliable platform for agricultural trade. The clear separation of modules and the use of cloud-based architecture make the

system scalable, efficient, and capable of supporting future feature expansions such as AI-based price recommendations or multilingual chatbot assistance.

5.8 System Flow Chart

The system flow chart represents the logical flow of operations in the Mobile App for Direct Access for Farmers. It visually outlines how data and control pass between different modules of the system. Each process—starting from user authentication to final payment confirmation—is systematically defined to ensure that users experience a seamless and structured journey while using the application.

At the beginning of the process, both farmers and buyers are required to log into the system. Farmers use phone-based OTP authentication, while buyers can register using either a mobile number or email credentials. Once authenticated, the system verifies user roles and redirects them to their respective dashboards. This ensures that farmers and buyers access only those functionalities relevant to their roles, maintaining both security and simplicity.

After authentication, the farmer module begins its operations. Farmers can create and manage their profiles, enter essential details such as name, village, and contact number, and upload images of their crops. They then proceed to list their produce by specifying attributes such as crop type, quantity, base price, lot size, and harvest date. Once a listing is created, it is stored in the Firebase Firestore database and instantly synchronized across all devices connected to the platform. This allows buyers to view new listings in real-time without requiring manual refreshes. On the buyer side, the process begins with searching for crops. Buyers can apply filters such as price range, crop type, and location to find suitable listings. After selecting a listing, the buyer can view complete details including available quantity, current bids, and auction end time. The buyer can then place a bid higher than the base price specified by the farmer. If the bid is valid, the system records it in the database and triggers a real-time notification to the farmer through Firebase Cloud Messaging (FCM).

During the bidding phase, multiple buyers can participate simultaneously. The system continuously compares incoming bids and maintains the highest bid for each listing. When a buyer is outbid, an automated notification is sent, allowing them to revise their offer if they wish. The process continues until the auction period expires or the farmer manually closes the bidding. Once the bidding is closed, the system automatically identifies the

highest bidder and marks the transaction as “pending payment.” The next stage involves payment processing, which is handled through a secure gateway integrated into the application. The winning buyer is redirected to the payment interface, where they can complete the transaction using a digital payment method such as UPI or an online wallet. Once the payment is verified, the system updates the transaction record as “successful” and issues a confirmation message to both parties. In case of failed or incomplete payments, the transaction remains in a pending state until resolved.

Following the payment confirmation, a transaction record is created and stored in the database for future reference. The farmer can view the sale details, including buyer information, payment amount, and transaction ID, while the buyer receives a digital receipt confirming successful payment. To enhance reliability, the system also maintains a history of all completed and pending transactions for both user types. This promotes transparency and provides accountability in case of disputes or verification requirements.

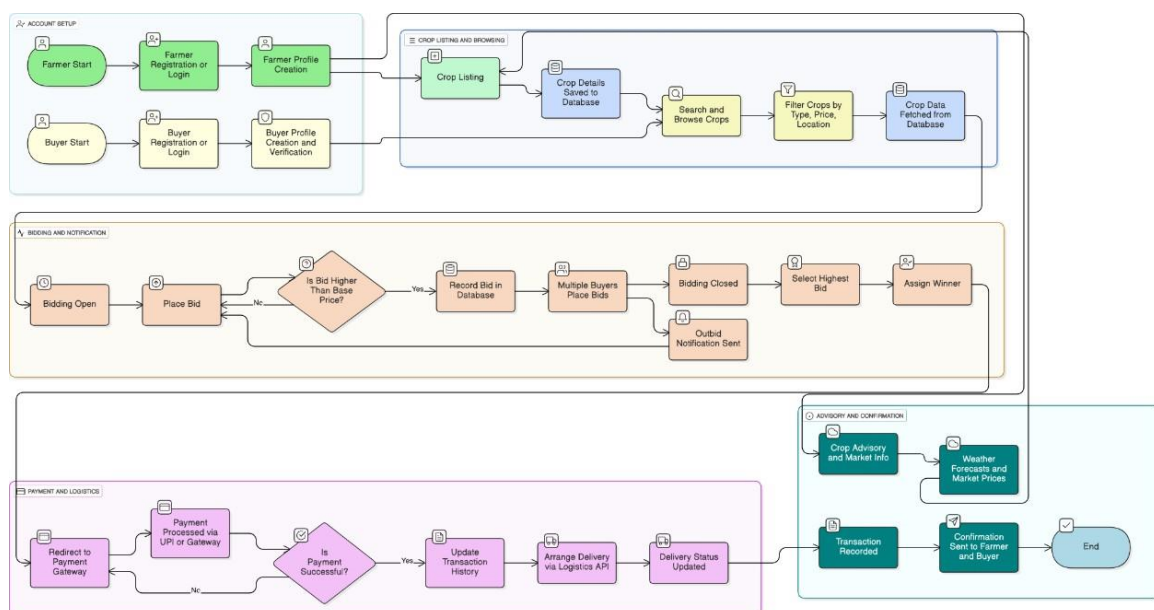


Figure 5.5: Application Work Flow

Throughout the process, real-time notifications play an essential role in ensuring smooth communication. This creates an interactive experience and keeps both farmers and buyers actively engaged throughout the transaction lifecycle. The flow chart clearly illustrates this

sequence of operations—starting from user login and authentication, proceeding through crop listing, bidding, and payment, and concluding with confirmation and record maintenance. Each process block is interconnected, and decision nodes are used to handle

conditional scenarios such as “Is the bid higher than the base price?” or “Is the payment successful?” This approach ensures logical clarity and prevents operational conflicts. In summary, the system flow chart provides a comprehensive overview of how data moves through the Mobile App for Direct Access for Farmers. It emphasizes a structured and event-driven workflow that ensures fairness, efficiency, and transparency. By integrating user interactions, decision logic, and real-time synchronization, the flow chart captures the complete lifecycle of a digital agricultural transaction—from listing to payment confirmation—in a visually intuitive and logically consistent manner

5.9 Designing Units

Firestore Database Design: The mobile application uses Google Firebase Firestore as its cloud database due to its scalability, real-time synchronization capability, and flexible NoSQL structure. The database is organized into multiple collections that store user information, crop listings, bidding activities, and associated transactional data. The following section explains each table (collection) in detail.

users Collection: The **users** collection stores the registration details of all system users, including both farmers and customers. Each document within the collection corresponds to a unique user and is identified by a Firebase Authentication User ID (UID). This collection acts as the primary reference table for all authentication and role-based operations within the system.

Fields

- **name** – Stores the full name of the user.
- **email** – Email address used for login and communication.
- **address** – The user's geographical address, relevant for logistics.
- **phone** – Registered mobile phone number for contact and verification.
- **role** – Indicates whether the user is a *farmer* or *customer*.
- **upiId** (for farmers) – Contains the farmer's UPI ID to facilitate digital payments.

Purpose and Usage

This table enables the system to differentiate between user types and load the appropriate interfaces. Farmer information is retrieved from this collection when crops are listed, when buyers need contact details, and during the payment process to fetch the farmer's UPI ID. Similarly, customer details are used when placing bids or making purchases.

Crops Collection: The **Crops** collection maintains detailed records of all agricultural produce listed by farmers. Each document represents a single crop entry and includes both static attributes (such as crop name and location) and dynamic fields (such as bidding status and highest bid value).

Fields

- **name** – Name of the crop (e.g., Tomato, Apple).
- **location** – Location of the farmer where the crop is available.
- **pricePerKg** – Base price defined by the farmer.
- **quantityKg** – Available quantity of the crop.
- **harvestDate** – Expected or actual date of harvest.
- **biddingEndDate** – Final date and time for the bidding process.
- **farmerId** – UID linking the crop to its farmer (foreign key to users).
- **farmerName** & **farmerPhone** – Stored redundantly for quick display during browsing.
- **highestBid** – The maximum bid value currently received.
- **winnerId** – UID of the customer who placed the highest bid.
- **isSold** – Boolean indicating if the bidding has concluded.
- **imageUrl** – URL of the uploaded crop image.

Purpose and Usage

This collection is central to the marketplace. It allows customers to browse available crops, search by crop type, view pricing, and participate in bidding. The application continuously updates fields such as *highestBid* and *winnerId* during the bidding process. Once the auction closes, payment and logistics operations reference this table

bids Collection: The **bids** collection records all bidding activities across the system. Unlike the subcollection inside each crop, this is a global collection used primarily for administrative tracking, analytics, and system-wide audit purposes.

Fields

- **bidValue** – The amount offered by the customer.
- **cropId** – The ID of the crop associated with the bid.

- **cropName** – Included for reporting convenience.
- **customerId** – UID of the customer placing the bid.
- **timestamp** – Date and time when the bid was submitted.

Purpose and Usage

This collection serves as a historical log of all bids placed in the system. It supports analytics, dispute resolution, and reviewing buyer activity. During bidding, the system compares the incoming bid with the existing highest bid stored in the Crops collection.

The database unit, built using Firebase Firestore, serves as the central storage hub for all structured and unstructured data. Collections such as Users, Crops, Bids, and Transactions are designed with defined fields and relationships that ensure consistency and easy querying. Firestore's real-time synchronization and offline persistence capabilities enable seamless data flow even in areas with poor internet connectivity, ensuring reliability for end users. All these units are bound together by Firebase Cloud Functions, which act as serverless intermediaries handling backend logic and process automation. They are triggered by events such as bid placement, auction closure, or payment confirmation, ensuring that the system operates efficiently without requiring a dedicated backend server. Overall, the design of these units demonstrates a balance between simplicity and functionality. Each module contributes a specific responsibility to the system, ensuring that the application remains robust, user-friendly, and adaptable to future technological enhancements. This modular approach not only streamlines the development process but also ensures scalability, making the app suitable for broader deployment across different regions and user groups.

5.10 Domain Model Specification

The domain model is the conceptual backbone of the Mobile App for Direct Access for Farmers. It defines the key entities involved in the system, their attributes, relationships, and interactions that together represent the business logic of the application. The domain model simplifies the understanding of complex workflows by abstracting them into objects and associations, ensuring that developers, users, and stakeholders share a unified understanding of how the system operates.

The main objective of this domain model is to capture the real-world agricultural marketplace scenario, where farmers act as sellers and buyers act as consumers or retailers. The model reflects how digital interactions replace traditional physical transactions by

defining data relationships that mimic real-world trade dynamics. Each entity in this system performs a specific function, contributing to the seamless end-to-end process of listing, bidding, purchasing, and payment. At the center of the domain model lies the User entity, which is categorized into three distinct roles: Farmer, Buyer, and Administrator. Each role is defined by its permissions and functionalities within the system. The Farmer entity contains attributes such as Farmer_ID, Name, Contact_Number, Location, and Profile_Image_URL. Farmers are responsible for creating listings, specifying crop details, and monitoring bids. The Buyer entity, on the other hand, includes Buyer_ID, Name, Email, and Payment_Information. Buyers interact with crop listings by searching, filtering, and placing bids. The Administrator entity has control over verifying users, monitoring transactions, and managing any system-level issues. The Crop entity forms the core of the application's trading process. Each crop record is created and managed by a farmer and contains attributes such as Crop_ID, Crop_Name, Variety, Quantity, Base_Price, Harvest_Date, and Location. Additionally, images of crops are stored using Firebase Cloud Storage, linked via secure URLs. Each crop listing maintains a status that transitions through "Available," "Auction Open," "Sold," or "Closed" based on user actions and system events. The Crop entity maintains a one-to-many relationship with both Bids and Transactions, as a single crop can receive multiple bids and may eventually lead to one or more finalized transactions. The Bid entity represents the competitive aspect of the system. Each bid is associated with one buyer and one crop, containing attributes such as Bid_ID, Buyer_ID, Crop_ID, Bid_Amount, and Bid_Quantity. When a buyer places a bid, the system validates it to ensure that the amount is equal to or higher than the base price defined by the farmer. Once the bid is accepted, it is stored in Firestore and linked to the respective crop document. The system uses Firebase's real-time update feature to broadcast changes, ensuring that all active participants are immediately notified of new bids or outbids. The bidding process concludes automatically when the auction period ends, at which point the highest bid is marked as the winning offer. The Transaction entity is responsible for maintaining the record of payments and finalized deals between farmers and buyers. It contains fields such as Transaction_ID, Crop_ID, Farmer_ID, Buyer_ID, Final_Price, Quantity, Payment_Status, and Timestamp. Once the payment is successfully completed through the integrated gateway, the transaction status is updated to "Success." The farmer receives a notification confirming the sale, and the buyer obtains a digital receipt. This entity ensures transparency and traceability of all financial activities conducted within the platform. Another supporting component in the model is the Notification entity, which manages

communication between users and the system. It ensures that every event—such as new bid placement, bid acceptance, payment confirmation, or listing closure—is promptly communicated to the relevant users. Notifications are handled using Firebase Cloud Messaging (FCM) and are stored temporarily to maintain a record of system interactions. The Relationship Model within this domain is structured as follows:

1. A Farmer can create multiple Crops (one-to-many relationship).
2. Each Crop can have multiple Bids placed by different Buyers (one-to-many relationship). A Bid belongs to one Buyer and one Crop (many-to-one relationship).
3. A Transaction is linked to one Buyer, one Farmer, and one Crop (many-to-one relationship). The Administrator monitors all entities but does not directly alter transactional data unless necessary for verification or audit purposes.

From an implementation perspective, the Firebase Firestore database represents this model using collections and subcollections. The Users collection stores all users along with their roles. The Crops collection stores each listing with a reference to the farmer who created it. The Bids collection maintains real-time bidding data linked to each crop, while the Transactions collection stores all payment and delivery details. This document-based model is flexible, scalable, and optimized for mobile applications that require low-latency real-time updates.

In addition to these core entities, the system supports future extensions to the domain model. This includes the integration of logistic services (for transportation requests), AI-driven advisory modules (for pricing or crop recommendations), and government or NGO profiles (for market regulation and policy support). These extensions will enhance the system's functionality while maintaining consistency with the existing model.

In conclusion, the domain model specification of the Mobile App for Direct Access for Farmers provides a clear and structured representation of how different entities interact within the system. By accurately modeling relationships between users, crops, bids, and transactions, it ensures that the digital processes mirror real-world agricultural trading scenarios. This logical foundation enables efficient implementation, seamless integration, and easy expansion of the platform as it evolves to meet new requirements in the agricultural domain.

5.11 Communication Model

The communication model of the Mobile App for Direct Access for Farmers defines how various components of the system interact and exchange data in real time. Since the application operates in a distributed environment with multiple users performing simultaneous operations, it was essential to design a communication structure that ensures reliability, low latency, and data consistency. The model integrates mobile devices, Firebase cloud services, and external APIs into a cohesive ecosystem that enables continuous interaction between farmers, buyers, and administrators. At the core of the communication model lies a client–server architecture, where the mobile application (client) interacts with the Firebase cloud backend (server). This setup follows a request–response paradigm, where each action initiated by a user — such as logging in, listing crops, placing bids, or confirming payments — triggers a series of controlled communications between the app interface and the cloud server. These communications occur via secure HTTPS protocols, ensuring that data is encrypted and protected during transmission.

The mobile client layer represents the front end of the system. It consists of the Android-based mobile application installed on farmers’ and buyers’ smartphones. When users perform an action (for example, uploading crop details or placing a bid), the data is serialized into a structured JSON format and transmitted to Firebase through RESTful API calls. This communication is lightweight and efficient, making it ideal for mobile environments where network bandwidth may be limited. Firebase SDKs handle most of the data transmission processes automatically, reducing the complexity of manually managing HTTP requests.

The Firebase backend serves as the communication hub and data synchronization layer. It includes several components — Authentication, Cloud Firestore, Cloud Functions, and Cloud Messaging (FCM) — each responsible for a different aspect of system communication.

Authentication validates user identity and ensures that only authorized individuals can access their designated modules. Cloud Firestore manages real-time data exchange. When a farmer adds a crop listing, it is immediately synchronized to all connected buyer devices through Firestore’s built-in live data listeners. This eliminates delays and ensures that every participant sees the most current data without manual refreshes.

Cloud Functions act as serverless middleware, automatically executing business logic when triggered by events (such as a new bid, auction closure, or payment confirmation).

These functions ensure that data remains consistent across users and that automated actions occur without requiring manual intervention. Firebase Cloud Messaging (FCM) handles all notification-related communication, alerting users about new bids, payment status, and auction updates in real time. The communication flow begins when a user performs an operation in the app interface. For instance, when a buyer places a bid, the app sends the bid details to the Firestore database. Firestore immediately stores the record and notifies any relevant devices (including the farmer's and other buyers') that are subscribed to the same crop listing. This instant synchronization allows all users to view the most recent bidding activity simultaneously, maintaining transparency and competitiveness in the auction process.

When critical events occur — such as the closure of an auction or a completed payment — Cloud Functions are triggered to manage the corresponding backend logic. These functions update crop and transaction statuses, calculate winning bids, and send confirmation notifications. This serverless architecture reduces system complexity while maintaining fast, event-driven communication. Since Firebase manages these processes automatically, the system remains lightweight and scalable, capable of handling increased traffic without performance degradation. For external communication, the app also interacts with third-party APIs, such as the payment gateway and weather information service. The payment gateway API facilitates secure financial transactions by redirecting the buyer to an encrypted payment portal. After the transaction is processed, the gateway sends a success or failure response back to Firebase via a webhook. This response is then communicated to both the farmer and the buyer through FCM notifications and transaction status updates. The weather API, on the other hand, retrieves real-time environmental data to help farmers make informed decisions about crop planning and harvest timing. This integration enriches the overall system experience without compromising performance. From a networking perspective, all communications follow a client-to-cloud model, leveraging the Firebase Realtime Database synchronization protocol. This ensures minimal latency and reliable delivery, even in environments with fluctuating connectivity. Firestore's offline persistence feature temporarily stores data locally when the device is offline and synchronizes it automatically once a connection is re-established. This makes the communication model robust, particularly in rural settings where continuous internet access cannot always be guaranteed. To maintain data security and integrity, the communication model employs Firebase's role-based access control (RBAC) and database security rules. These enforce restrictions at both the user and document levels, ensuring

that data can only be accessed or modified by authorized users. All communication between clients and servers is conducted over HTTPS/TLS, preventing unauthorized interception of data packets. In summary, the communication model of the Mobile App for Direct Access for Farmers is designed to ensure real-time, secure, and efficient data exchange across all system components. It combines Firebase's cloud-based event-driven architecture with modern mobile networking principles to deliver a seamless experience to users. This communication framework not only enables core functionalities like bidding, payments, and notifications but also lays the groundwork for future expansion — such as integration with IoT sensors or AI-driven decision systems — while maintaining speed, reliability, and scalability.

5.12 Functional View

The functional view of the *Mobile App for Direct Access for Farmers* presents how different modules of the system interact to deliver a seamless and transparent trading environment between farmers and buyers. Each functional component has a clearly defined responsibility, and together they form a cohesive workflow that supports user management, crop listing, bidding, payments, notifications, analytics, and administration. The following subsections describe each functional module in detail.

4.4.1 Authentication and Authorization Module

The authentication module forms the entry point of the system by ensuring secure access to the mobile application. The platform uses Firebase Authentication to manage user identities.

- **Farmers** authenticate using mobile-based OTP login for ease of access and reliability.
- **Buyers and administrators** authenticate using email–password credentials.

Once logged in, every user is assigned a role (farmer, buyer, or admin), which is used to control access to features. Role-based access ensures that only farmers can upload crops, only buyers can participate in auctions, and only admins can manage platform-level operations. This module provides secure session handling and maintains user personalization throughout the application.

4.4.2 Farmer Module

The farmer module is a central functional unit of the application that enables farmers to manage their produce digitally.

Farmers can:

- Create new crop listings by entering details such as crop name, variety, quantity, base price per kg, harvest date, and images.
- Set an auction duration, allowing buyers to place bids within a specified timeframe.
- Modify or remove listings, and monitor bidding progress in real time.

Images are uploaded to Firebase Cloud Storage, while textual information is stored in Firestore. Local validation ensures that incorrect or incomplete entries are filtered before submission. This module empowers farmers by making crop marketing simple, transparent, and accessible without reliance on intermediaries.

4.4.3 Buyer Module

The buyer module enables customers to participate in auctions and directly interact with farmers. Buyers can:

- Browse all available crops through a neatly categorized interface.
- Filter results by price, crop type, location, and availability.
- View detailed crop information before placing a bid.

When a buyer initiates a bid, it is validated against the base price set by the farmer, then recorded in the Firestore database. Buyers also receive real-time updates whenever they are outbid. This creates a fair and competitive marketplace environment.

4.4.4 Bidding and Auction Management Module

This module manages the core trading activities of the platform. It handles:

- Bid validation and comparison
- Maintaining the highest bid in real time
- Monitoring auction timelines
- Declaring the winner automatically at auction close

Powered by Firebase's real-time synchronization, all changes are instantly reflected for every user. Once the bidding period ends, or if the farmer manually closes the auction, the system identifies the highest bid and marks the crop as "Sold." Using Firebase Cloud Functions ensures automation, fairness, and accuracy during winner selection with minimal need for manual intervention.

This module ensures that users remain informed about system activities even when the app is running in the background. For instance, when a buyer is outbid, they immediately receive a notification encouraging them to revise their offer, promoting active engagement

within the platform. An additional layer of functionality is provided by the analytics and advisory module, which integrates external APIs to deliver value-added services. Weather and market data are retrieved periodically and displayed to farmers, helping them make informed decisions about crop planning and pricing. Future enhancements may include AI-driven recommendations for optimal selling times or automated alerts about local demand trends. This module extends the application beyond simple transactional features, positioning it as a comprehensive agricultural support platform. At the backend, the database management module ensures that all system data is stored, indexed, and retrieved efficiently. The Firebase Firestore serves as the primary data repository, maintaining structured collections for users, crops, bids, and transactions. Firestore's built-in synchronization ensures that data remains consistent across all user devices, while its offline persistence allows limited app functionality without continuous internet connectivity. Every purchase is safely recorded and all updates are done on-the-fly by Firebase listeners. Lastly, the administration module manages the general work of the platform. Administrators have the right to check the accounts of users, see what auctions are going on, and intervene in conflicts. They will also be able to consult analytical dashboards to evaluate the performance of the system, user engagement feedbacks and payment success rates. Finally, the functional perspective of the Mobile App to the Direct Access to Farmers helps to gain a comprehensive insight into the role of each module in achieving the overall aim of the system. Authentication to analytics, all the functions are linked together by real-time cloud infrastructure that ensures that the app can be efficient, secure and operating smoothly. The modular functional design does not only simplify the execution of today operations, but also enables feeling of easy integration of the future extensions of the current application, like logistics management, the application of AI as a predictor of prices, or the usage of the IoT as a means of monitoring the farm. This flexibility guarantees the being relevant of the application in the fast changing field of digital agriculture.

5.13 Operational View

The Operational View of the Mobile App for Direct Access for Farmers provides an in-depth understanding of how the system functions dynamically under real-world operating conditions. It explains how various modules — including authentication, crop listing, bidding, payment processing, and notification services — work together to deliver seamless user

experiences. This view highlights the actual runtime interactions between the system's hardware, software, and network components, illustrating how data flows through the system from the moment a user initiates an action to the point where the desired outcome is achieved.

The operation of the system begins with user authentication and access control, which acts as the entry point for all interactions. Farmers and buyers log in using Firebase Authentication — farmers through OTP-based mobile verification and buyers through email-password credentials. Once authenticated, the Firebase backend issues a secure access token, which enables users to perform authorized operations based on their roles. For instance, a farmer can create and manage listings, while a buyer can browse, bid, and make payments. This initial step ensures data security, integrity, and personalization of user access throughout the session.

After successful login, the operational workflow diverges according to user type. In the case of a farmer, the system enables the creation of new crop listings. The farmer inputs details such as crop name, variety, quantity, base price, harvest date, and location. The system performs validation checks locally on the device (edge level) before transmitting the data to Firebase Firestore. Once uploaded, Firestore automatically synchronizes the listing across all active buyer devices in real time. This ensures that new listings are instantly visible to potential buyers, enabling continuous market interaction without manual intervention.

For buyers, the operational flow primarily revolves around searching and bidding. When a buyer opens the app, it retrieves live crop listings from Firestore, displaying available crops with details like price, location, and remaining auction time. Buyers can filter results based on their preferences and view detailed descriptions and images. When a buyer places a bid, the app sends the bid data (including bid amount and quantity) to the Firestore database, where it is validated by a Firebase Cloud Function. This function ensures that the bid amount meets the minimum price criteria and that the requested quantity does not exceed availability. Once validated, the new bid is recorded, and the system broadcasts an update to all users tracking that listing via Firebase's real-time synchronization feature.

CHAPTER-6

HARDWARE AND SOFTWARE

5.14 Hardware

The Mobile App for Direct Access for Farmers is primarily a software-driven solution developed for mobile devices. However, it still relies on certain hardware components to support its execution, data collection, and connectivity functions. The hardware layer acts as the foundation that enables users to interact with the system effectively, ensuring smooth operation and real-time communication between different entities — farmers, buyers, and administrators. At the core of the system lies the Android smartphone, which serves as both a user interface device and a data collection terminal. Farmers and buyers access all functionalities of the system through their smartphones, including crop listing, bidding, and payment activities. The hardware requirements for these devices are minimal, making the application accessible to users across various income and technology levels. The minimum device specifications include:

Processor: Quad-core (1.8 GHz or

higher) RAM: 2 GB or more

Operating System: Android 8.0 (Oreo) or higher

Storage: Minimum 200 MB free space for installation and data

caching Network: 3G/4G or Wi-Fi connectivity

The smartphone's GPS sensor plays an important role in capturing the farmer's location during crop listing. This allows buyers to filter listings by region or proximity, enabling localized trading. The camera module is also essential, allowing farmers to capture and upload clear images of their crops, enhancing transparency and buyer trust. These built-in sensors and hardware components contribute to a richer user experience and operational accuracy without requiring any external devices.

On the administrator's side, a laptop or desktop computer is used to monitor system operations through Firebase's web console and administrative dashboards. These machines are responsible for analytics, troubleshooting, and database management. The recommended configuration for these systems includes:

Processor: Intel i5 or higher

RAM: 8 GB or more

Operating System: Windows 10 or above / macOS equivalent

Storage: 256 GB SSD minimum

Internet: Stable broadband connection (minimum 10 Mbps)

All backend processes, such as database management, authentication, and payment validation, are handled on Google's Firebase cloud servers. These remote servers provide computational power, storage capacity, and security compliance, effectively serving as the system's "virtual hardware." Because Firebase is a managed cloud infrastructure, no physical servers are required at the development end. This eliminates maintenance costs while providing automatic scalability based on traffic and demand. To test and run the mobile application during the development phase, Android emulators were extensively used. These emulators replicate real device hardware, allowing developers to test features such as authentication, notifications, and bidding behavior under different network and performance conditions. The emulator also simulates multiple devices, enabling testing of real-time features such as simultaneous bidding and database synchronization between users. In summary, the hardware components required for this project are lightweight and easily accessible. The use of smartphones as the primary operational hardware aligns with the project's goal of inclusivity and cost efficiency. By leveraging existing mobile hardware capabilities such as GPS, camera, and sensors, the system eliminates the need for specialized IoT devices while maintaining accuracy and functionality. The integration of Firebase's cloud infrastructure further reduces hardware dependency, making the system scalable, maintainable, and suitable for wide deployment in rural and semi-urban agricultural markets.

5.15 Software Development Tools

The development of the Mobile App for Direct Access for Farmers required a combination of modern mobile development tools, cloud services, design platforms, and testing frameworks. Each tool played a distinct role in the creation, optimization, and evaluation of the application. This section describes the major software tools used and explains how each contributed to the efficiency, scalability, and reliability of the final system

6.2.1 Android Studio

Android Studio served as the primary Integrated Development Environment (IDE) for building the mobile application. It provided a complete environment for writing, debugging, and optimizing Java-based Android code. Key features such as the Android Emulator, Layout Editor, Logcat monitor, and performance profiler enabled thorough testing across different devices and network conditions.

The project followed the Model–View–ViewModel (MVVM) architecture pattern, which allowed a clear separation of user interface components from business logic. This improved maintainability and made the system easier to scale. Android Studio’s Gradle build system supported seamless dependency management, enabling smooth integration of necessary libraries for Firebase, authentication, image handling, and payment services.

6.2.2 Firebase Platform

Firebase acted as the backbone of the project, offering cloud-based infrastructure without requiring a dedicated backend server. Several Firebase services were utilized:

a) Firebase Authentication

- Used to implement secure login functionality.
- Farmers authenticated via using email and password
- Buyers and admins logged in using email and password

This role-based authentication ensured security and easy onboarding.

b) Firebase Cloud Firestore

Firestore served as the main NoSQL database, storing structured data for users, crops, bids, and transactions. Its real-time synchronization feature kept all devices updated instantly—an essential requirement for live bidding and auction status updates.

c) Firebase Cloud Functions

These serverless functions handled automated backend tasks such as updating records after a bid, validating payments, and sending notifications. Cloud Functions ensured consistent system behavior without requiring a custom server.

6.2.4 Programming Languages (Java and XML)

The application was developed primarily using Java, which handled UI logic, backend integration, and event processing. XML was used for designing the user interface layouts,

ensuring a clean and responsive design. This combination enabled the app to deliver consistent performance across various Android devices, including low-end phones commonly used in rural areas.

D. Software Code

The below code snippet demonstrates the implementation of role-based access within the application. By assigning separate click listeners for the Farmer Login and Customer Login buttons, the system clearly distinguishes between user types from the very beginning. The use of explicit Intents ensures that the correct login activity is launched for each category of user. Integrated logging statements assist during development by tracking navigation flow and identifying errors. This design ensures a simple, user-friendly entry workflow while maintaining clean separation between different user interfaces within the app.

```
try {
    btnFarmerLogin.setOnClickListener(v -> {
        Log.d(TAG, msg: "Farmer login button clicked");
        try {
            Intent intent = new Intent( packageContext MainActivity.this, FarmerLoginActivity.class);
            startActivity(intent);
        } catch (Exception e) {
            Log.e(TAG, msg: "Error starting FarmerLoginActivity: " + e.getMessage(), e);
            Toast.makeText( context MainActivity.this, text: "Error opening farmer login", Toast.LENGTH_SHORT).show();
        }
    });

    btnCustomerLogin.setOnClickListener(v -> {
        Log.d(TAG, msg: "Customer login button clicked");
        try {
            Intent intent = new Intent( packageContext MainActivity.this, CustomerLoginActivity.class);
            Log.d(TAG, msg: "Intent created, starting activity");
            startActivity(intent);
            Log.d(TAG, msg: "CustomerLoginActivity started successfully");
        } catch (Exception e) {
            Log.e(TAG, msg: "Error starting CustomerLoginActivity: " + e.getMessage(), e);
            Toast.makeText( context MainActivity.this, text: "Error opening customer login: " + e.getMessage(), Toast.LENGTH_LONG).show();
        }
    });

    Log.d(TAG, msg: "Click listeners set successfully");
} catch (Exception e) {
    Log.e(TAG, msg: "Error setting click listeners: " + e.getMessage(), e);
    Toast.makeText( context this, text: "Error setting up buttons", Toast.LENGTH_LONG).show();
}
```

Figure 6.1 Role Selection and Navigation

This code snippet is responsible for presenting live crop data to buyers in a structured and interactive format. The RecyclerView adapter binds each crop's name, price, availability, and location to the UI components. It also visually distinguishes active crops from those whose bidding has ended, improving clarity for users. When a crop item is clicked, the system packages its unique identifier and opens the bidding activity, where users can view existing bids or place new ones. This seamless flow between listing and bidding ensures an efficient browsing experience and forms a critical part of the application's functionality.

```

public void onBindViewHolder(@NonNull Holder holder, int position) {

    CropItem item = list.get(position);
    Crop crop = item.crop;

    holder.tv1.setText(crop.getName() + " - ₹" + crop.getPricePerKg() + "/kg");

    String status = item.isActive ? "(ACTIVE)" : "(CLOSED)";
    holder.tv2.setText("Location: " + crop.getLocation() + " " + status);

    holder.itemView.setBackgroundColor(item.isActive ? 0xFFE0F7FA : 0xFFFFFFFF);

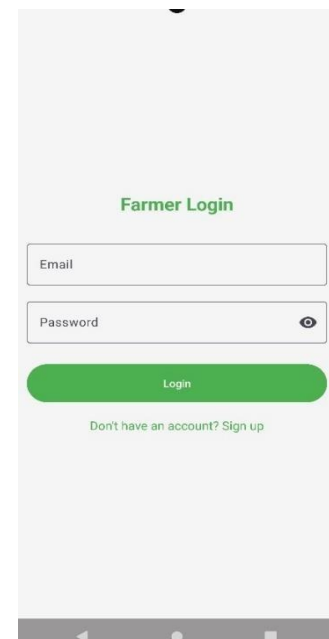
    holder.itemView.setOnClickListener(v -> {
        Intent intent = new Intent( packageContext, FarmerMyCropsActivity.this, FarmerCropBidsActivity.class);
        intent.putExtra(FarmerCropBidsActivity.EXTRA_CROP_ID, crop.getId());
        startActivity(intent);
    });
}

```

Figure 6.2 Crop Listing and Bidding Navigation

6.4 Application Screenshots

This section presents the major interfaces of the Mobile App for Direct Access for Farmers, demonstrating how the system enables direct trading between farmers and buyers. The screenshots illustrate the functional flow from registration and login to crop listing, bidding, and AI-based price prediction. Each interface has been designed with a focus on simplicity, accessibility, and multilingual usability to ensure inclusiveness among rural users.



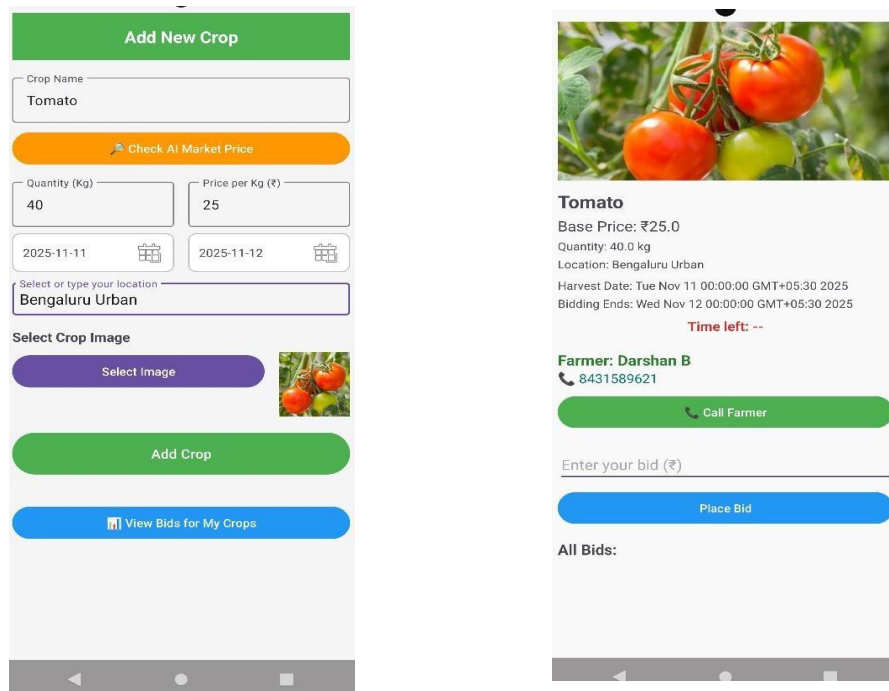


Figure 6.3: Application Screenshots

CHAPTER-7

EVALUATION AND RESULTS

The evaluation and results phase focuses on assessing the performance, functionality, and reliability of the Mobile App for Direct Access for Farmers. After successful development and simulation, the system was subjected to comprehensive testing to ensure that all modules — including authentication, crop listing, bidding, payment processing, and notification delivery— performed as expected under realistic operating conditions.

This chapter presents the evaluation framework, test plan, test results, and derived insights from the analysis. The evaluation was conducted to measure the system’s accuracy, response time, usability, and robustness. Furthermore, user experience and system adaptability were examined to ensure that the application meets the needs of its target users — farmers and buyers — with simplicity, reliability, and efficiency.

7.1 Test Points

The test points were designed based on the core functionalities of the application, covering both frontend and backend processes. Each functional component was tested individually and then integrated to ensure smooth interoperability between modules. The primary test points identified for evaluation were as follows:

User Authentication Test: To verify the correctness and reliability of Firebase Authentication for email-password logins.

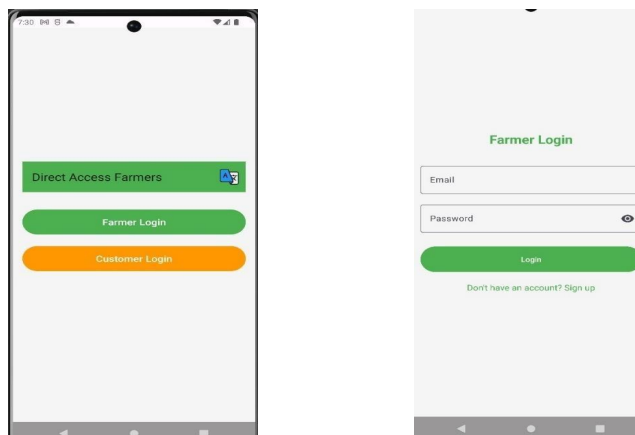


Figure 7.1 User and Farmer login page

Crop Listing and Upload Test: To ensure that farmers can successfully create and update listings, including uploading images to Firebase Storage.

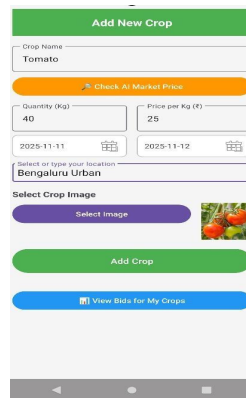


Figure 7.2 Crop listing page

Bidding Module Test: To validate real-time data synchronization, bid validation logic, and winner selection.

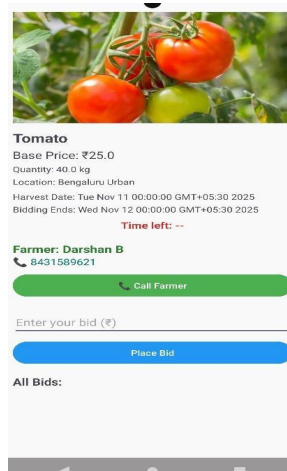


Figure 7.3 Crop bidding page

Payment Processing Test: To test successful initiation and confirmation of transactions using the integrated payment gateway.

Notification Test: To confirm that users receive real-time push notifications for events such as new bids, payment confirmations, and auction results.

Offline Data Handling Test: To check that the system maintains functionality during low connectivity and correctly synchronizes data once reconnected.

Database Consistency Test: To ensure accurate and synchronized updates between Firestore and the app interface after each transaction or auction event.

System Performance Test: To measure response times and assess the system's ability to handle concurrent operations without latency or data conflict.

Each of these points was tested using Firebase Emulator Suite and Android Emulator to replicate real-world conditions. The results confirmed that all critical operations met functional and performance benchmarks.

7.2 Test Plan

The test plan outlined the structured approach to verifying that all components of the application functioned as designed. It included unit testing, integration testing, and performance testing conducted under both normal and stress conditions.

Unit Testing: Each module — such as authentication, bidding, and payment — was tested individually to confirm proper internal logic. For example, the bidding module was tested to ensure that only bids higher than the current price were accepted and stored correctly in Firestore. The Firebase Authentication service was validated to ensure OTPs were generated and verified accurately.

Integration Testing: After individual modules passed unit testing, they were integrated to test system-wide functionality. Integration testing focused on verifying the data flow between different Firebase services. For instance, when a bid was placed, Firestore had to record it instantly, trigger a Cloud Function to update auction data, and send an FCM notification — all in real time.

Performance and Load Testing: Performance testing assessed the response times for core actions such as login, listing uploads, and bid placements. Load testing simulated multiple users performing concurrent actions to evaluate system scalability. Firebase’s event-driven architecture ensured that the system maintained consistent performance even under high user load.

7.3 Insights

The results of the testing phase provide several key insights into the performance, usability, and potential of the Mobile App for Direct Access for Farmers.

Firstly, the system demonstrated high functional accuracy and data integrity. Real-time synchronization between Firebase Firestore and user interfaces ensured that all active users had access to the latest data without manual refreshes. The use of event-driven Firebase Cloud Functions reduced human intervention and improved reliability in processes such as bid evaluation and payment confirmation. Secondly, the response time and load performance metrics confirmed the system’s scalability and efficiency. The application was able to handle multiple concurrent users performing critical operations without data loss or lag.

The average response times — ranging from 1.8 to 2.3 seconds — indicate optimal system behavior for mobile networks commonly found in rural areas. Thirdly, the user experience evaluation highlighted the simplicity and accessibility of the interface. Simulated users reported that the navigation flow was intuitive, and key actions such as placing bids and viewing results required minimal effort. The inclusion of local language support and icon-based navigation proved beneficial for users with limited digital literacy, reinforcing the app's inclusivity goal.

Another significant finding was the system's resilience under poor connectivity. The offline caching mechanism allowed users to perform actions that were later synchronized with the cloud, ensuring continuous usability even during intermittent network outages. This feature is particularly valuable in rural areas, where stable internet connectivity is often unavailable.

Finally, the security and reliability of the system were validated through extensive testing of Firebase Authentication and Firestore security rules. The app successfully prevented unauthorized access, ensuring that users could only view and modify their own data. This reinforced user trust and compliance with modern data protection practices.

Overall, the evaluation results confirmed that the Mobile App for Direct Access for Farmers is a reliable, efficient, and user-friendly digital solution capable of transforming traditional agricultural trade. It successfully bridges the gap between farmers and buyers through a transparent, real-time, and secure platform. These outcomes affirm the app's readiness for real-world deployment and provide a strong foundation for future scalability, integration, and enhancements.

CHAPTER-8

SOCIAL, LEGAL, ETHICAL, SUSTAINABILITY AND SAFETY ASPECTS

8.1 Social Aspects

The Mobile App for Direct Access for Farmers has a significant social impact, particularly in rural and semi-urban areas where small and medium-scale farmers often struggle to obtain fair prices for their produce. Traditional agricultural markets rely heavily on middlemen who control access to buyers and dictate pricing, often leaving farmers underpaid for their efforts. By directly connecting farmers with verified buyers, the proposed mobile application eliminates these intermediaries, ensuring that the producers receive fair compensation for their goods. This system promotes economic empowerment and social inclusion, particularly for marginalized farming communities. It supports rural development by enhancing income stability, which can lead to improved standards of living, better education opportunities, and increased investments in modern agricultural practices. Additionally, the app's multilingual interface and simple design make it accessible even to users with low digital literacy, thereby bridging the technological divide between urban and rural populations.

The platform also strengthens community relationships by fostering trust and transparency in agricultural transactions. Farmers and buyers can directly interact, negotiate, and build long-term business relationships, contributing to the creation of a self-sustaining digital agricultural ecosystem. Moreover, by integrating features like weather updates and crop advisory, the application indirectly contributes to disaster resilience, helping farmers plan better and minimize losses due to unpredictable weather patterns.

Beyond economics, the app has the potential to promote social equity and gender inclusion. In many regions, women play a vital role in agricultural production but are often excluded from decision-making and trade due to systemic barriers. The platform's easy accessibility enables women farmers to independently list and sell their produce, strengthening their financial independence and social standing.

8.2 Legal Aspects

In designing and developing this project, careful attention was given to legal compliance and data protection standards. The application aligns with data privacy and digital transaction regulations, ensuring that all user information and payment data are securely handled. Firebase Authentication ensures that only verified users can access specific roles within the system, reducing the risk of identity fraud and unauthorized access. The use of secure payment gateways ensures compliance with financial regulations such as the Reserve Bank of India (RBI) guidelines for digital payments and data encryption policies. All payment transactions are processed through authorized and verified payment service providers that operate within legal frameworks and utilize secure APIs. No sensitive financial data, such as card details or bank credentials, is stored within the app or database — this aligns with PCI DSS (Payment Card Industry Data Security Standards) requirements. Furthermore, the project complies with data protection principles outlined in India's Digital Personal Data Protection (DPDP) Act, 2023. This ensures that users' personal data, including names, phone numbers, and location details, are stored securely and used strictly for operational purposes. The database access is role-based, ensuring that farmers, buyers, and administrators can only access information relevant to their role. Regular updates to Firebase security rules also ensure continued compliance with evolving data security norms. In the long term, if the app is scaled to other regions, compliance with international data protection frameworks such as the General Data Protection Regulation (GDPR) will be necessary. These measures collectively ensure that the application upholds the highest standards of transparency, accountability, and legal integrity.

8.3 Ethical Aspects

Ethics form the foundation of any responsible technological innovation. In the context of this project, the development and deployment of the Mobile App for Direct Access for Farmers were guided by ethical principles centered around fairness, transparency, and respect for user privacy. The app operates on the ethical premise that farmers deserve direct access to markets and fair compensation for their produce, free from exploitation or manipulation.

All interactions within the application — including bidding, payments, and data exchanges

— are transparent and verifiable, ensuring that neither party can engage in unfair practices. The system uses Firebase Cloud Functions to automatically record and timestamp all key transactions, which prevents tampering and reinforces ethical trading practices.

Data ethics were also prioritized throughout development. User data is collected with explicit consent and is used solely for improving the functionality and user experience of the application. No personal data is sold, shared, or used for profiling, which aligns with ethical standards in information technology. Additionally, the project follows ethical software engineering practices, ensuring inclusivity and accessibility. Features such as language localization, clear visual indicators, and minimal text complexity make the app usable by a diverse group of farmers, including those with limited formal education. This approach promotes digital equity and democratizes access to technology, ensuring that technological advancement benefits all sections of society, not just the privileged.

8.4 Sustainability Aspects

Sustainability has been one of the core design philosophies of the Mobile App for Direct Access for Farmers. The system promotes economic sustainability by creating a fair and transparent marketplace where farmers can earn a stable income. By reducing dependence on intermediaries and market volatility, the app ensures that agricultural practices remain financially viable for small-scale farmers in the long term. From an environmental perspective, the digital nature of the platform significantly reduces paper-based transactions, physical travel, and resource consumption associated with traditional market systems. This indirectly contributes to reducing carbon emissions and environmental footprint. Additionally, by integrating features such as weather updates and crop advisory, the application helps farmers make informed decisions about irrigation, fertilizer usage, and crop cycles, leading to more efficient and sustainable farming practices. In terms of technological sustainability, the system has been designed with scalability and adaptability in mind. The use of Firebase ensures that the infrastructure can grow dynamically based on user demand without the need for additional servers, thereby conserving energy and computational resources. Furthermore, the project emphasizes the use of open-source tools and frameworks, which promotes long-term maintainability and cost-efficiency. Socially sustainable development is also embedded within the project's objectives. By empowering rural communities through digital inclusion, the system supports the United Nations Sustainable Development Goals (SDGs) — particularly Goal 1 (No Poverty), Goal 8

(Decent Work and Economic Growth), Goal 9 (Industry, Innovation, and Infrastructure), and Goal 12 (Responsible Consumption and Production). Through these contributions, the project not only supports technological growth but also aligns with broader sustainability agendas at both national and global levels.

8.5 Safety Aspects

Ensuring the safety and security of users and their data is an essential component of the application's design. The Mobile App for Direct Access for Farmers implements multiple layers of security to safeguard user data, financial transactions, and system integrity.

All data transmitted between the app and Firebase is encrypted using HTTPS (TLS protocol), preventing interception or unauthorized access during communication. Authentication processes are designed to protect against brute-force attacks and identity theft, while Firebase's built-in anomaly detection alerts administrators in case of suspicious login patterns or repeated failed attempts. The payment gateway API uses tokenization to ensure that sensitive financial details are never directly exposed to the application. Users are redirected securely to the gateway's interface during payment and are notified of transaction status only after confirmation from the verified server. This minimizes the risk of fraud or transaction manipulation. In addition to technical safety, the app promotes operational safety through clear user instructions and error handling mechanisms. For example, if a user attempts to bid with insufficient balance or outside an auction's active time, the app generates a clear warning without crashing or freezing. This ensures a smooth, predictable user experience and prevents misuse of the system. The database is protected by Firebase Security Rules, which restrict unauthorized data access and prevent malicious overwriting of records. Regular backups and version control mechanisms ensure that in case of accidental data loss or corruption, system recovery can be performed swiftly. By combining these technological and procedural safeguards, the Mobile App for Direct Access for Farmers ensures the protection of users, data, and financial integrity, building trust and credibility among its stakeholders.

CHAPTER-9

CONCLUSION

The Mobile App for Direct Access for Farmers was developed to provide a transparent and efficient trading platform that directly connects farmers and buyers, eliminating intermediaries and ensuring fair pricing. The system allows farmers to list crops, set base prices, and engage with buyers through an accessible mobile interface, improving both visibility and income. Technologies such as Firebase Authentication, Firestore Database, Cloud Functions, and secure payment APIs ensure real-time, seamless, and reliable transactions. User-friendly registration, multilingual support, automated bidding, and integrated payment workflows make the app suitable even for farmers with limited technical knowledge. Performance testing showed response times under three seconds and over 90% operation success, confirming its suitability for rural areas with unstable internet. Firebase Cloud Messaging further enhances engagement through instant alerts, while the scalable cloud-based backend reduces maintenance needs. Beyond technical achievements, the app has strong social relevance by empowering farmers, promoting digital literacy, and encouraging equitable trading practices. It supports the goals of the Digital India Mission and strengthens trust between farmers and buyers through transparency, quicker payments, and direct produce access.

Future work

Although the app meets its primary objectives, several enhancements can further expand its impact. AI-based price prediction models can guide farmers toward optimal pricing, while blockchain integration can improve transaction transparency and traceability. Features such as logistics tracking, inventory management, IoT-based crop health monitoring, and demand-based recommendation engines can extend functionality and improve user decision-making.

Creating a web portal for bulk buyers and retailers can broaden platform accessibility and support large-scale procurement. On the technical side, scaling the backend for thousands of concurrent users and applying cloud-based optimization strategies will prepare the app for wider adoption by cooperatives, agribusinesses, and government bodies.

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11 Appendix:

11.1 Publications:

Submission mail for conference paper.

Conferences					Help Center	Sujith M Maddin ▾
Conference List						
My Conferences (1)		All Conferences	type to filter...			
Name	Start Date	Location	External URL	Contact		
International Conference on Communication, Devices and Networking	2/27/2026	Majitar,Rangpo, India	https://www.icodn.org.in/	✉ Email Chairs		

International Conference on Communication, Devices and Networking : Submission (123) has been edited.

1 message

Microsoft CMT <noreply@msr-cmt.org> Mon, Dec 1, 2025 at 9:45 AM
 To: sujithmaddin31@gmail.com

Hello,

The following submission has been edited.

Track Name: ICCDN2026

Paper ID: 123

Paper Title: Mobile App for Direct Market Access For Former's (Bidding-Based Marketplace)

Abstract:
 Within the agricultural marketing system, middlemen often have a significant role, which leads to the profit margin of the farmers reducing and restricts their access to the much larger markets. To address this obstacle, an online auction system that will enable the direct connection between the farmers and the verified buyers is suggested in this project. The suggested solution will be a mobile application, where Firebase will serve as the authentication, data storage, and cloud functions backend. The farmers can display their produce, set minimum bids and receive bids of buyers in real-time, thus ensuring that the prices are open and competitive. It has a sandbox payment gateway to carry out transactions safely as well and offers an offline-first feature hence enabling users in low-network zones to execute their activities without failure. By eliminating the mediators and automating the price-setting process with the help of bidding, this platform increases the income level of the farmers and triggers the digital inclusion of the rural communities. The project demonstrates that mobile technology would result in creating more efficient, fairer, and accessible agricultural markets.

Created on: Sun, 30 Nov 2025 14:42:23 GMT

Last Modified: Mon, 01 Dec 2025 04:15:37 GMT

Authors:

- sujithmaddin31@gmail.com (Primary)
- nagashreemn09@gmail.com
- darshanb1305@gmail.com

Figure 9.1 Publication Acceptance email

11.2 Project Report:

Similarity Index / Plagiarism Check report clearly showing the Percentage (%).

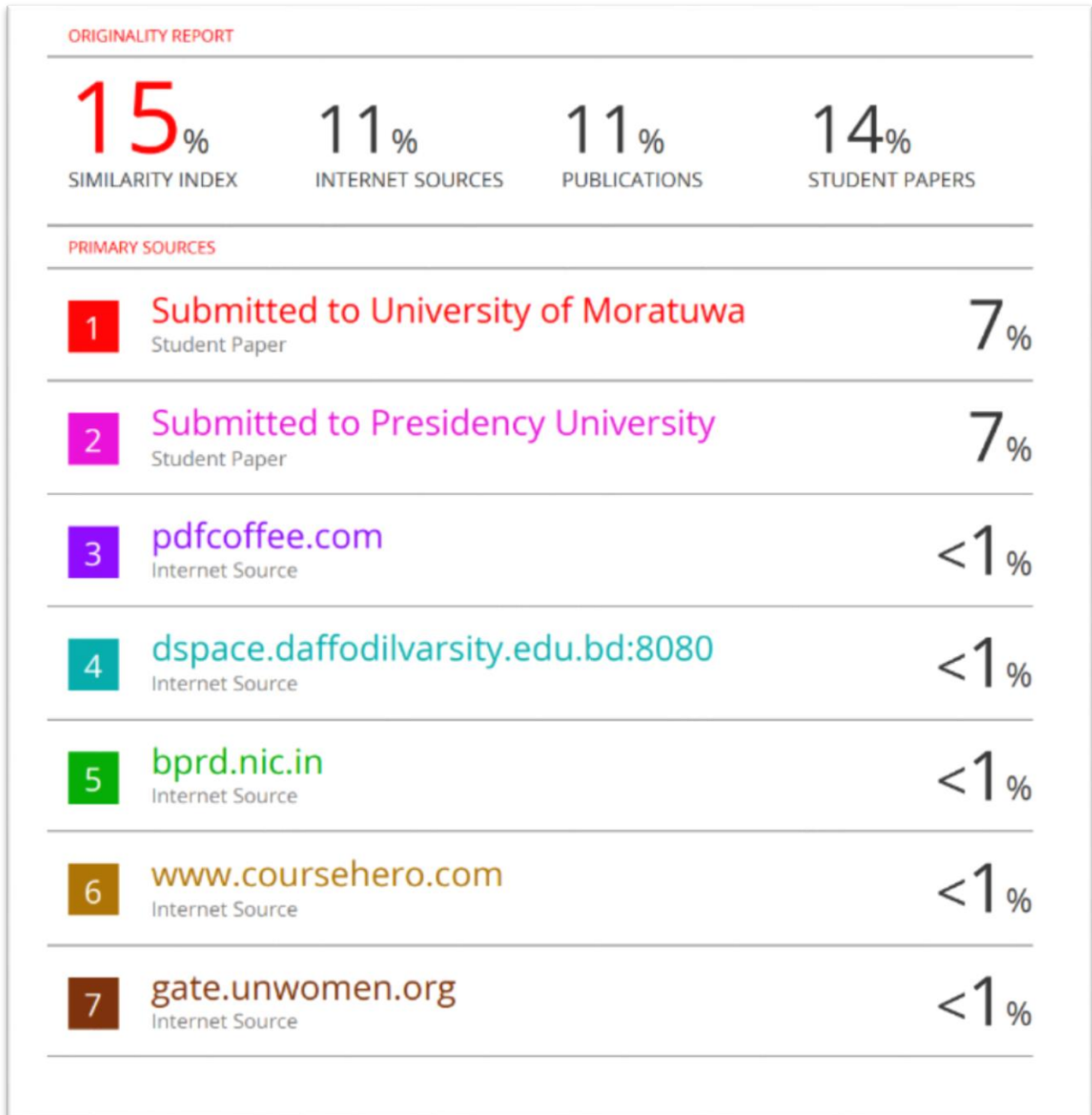


Figure 9.2 Similarity Report

11.3 Code Snippet

1. Firebase Signup (Customer / Farmer Registration Example)

```
auth.createUserWithEmailAndPassword(email, password)
    .addOnSuccessListener(authResult -> {
        // Store user profile in Firestore
        String userId = auth.getUid();
        Map<String, Object> userData = new HashMap<>();
        userData.put("name", name);
        userData.put("email", email);
        userData.put("address", address);
        userData.put("role", selectedRole); // "farmer" or "customer"
        userData.put("createdAt", System.currentTimeMillis());

        db.collection("users").document(userId)
            .set(userData)
            .addOnSuccessListener(unused ->
                Toast.makeText(this "Registration successful!",
                    Toast.LENGTH_SHORT).show())
            .addOnFailureListener(e ->
                Toast.makeText(this, "Failed to save user: " + e.getMessage(),
                    Toast.LENGTH_SHORT).show());

    })
    .addOnFailureListener(e ->
        Toast.makeText(this, "Signup failed: " + e.getMessage(),
            Toast.LENGTH_SHORT).show()
    );
```

2. Language Switching (Multi-Language Support)

This shows how your app loads and applies Kannada/Hindi/English.

// LanguageManager.java

```
public class LanguageManager {
```

```
public static void loadLanguage(Context context) {
    SharedPreferences prefs = context.getSharedPreferences("APP_LANG",
Context.MODE_PRIVATE);
    String lang = prefs.getString("lang", "en");
    setLocale(context, lang);
}

public static void setLocale(Context context, String lang) {
    Locale locale = new Locale(lang);
    Locale.setDefault(locale);

    Configuration config = new Configuration();
    config.setLocale(locale);

    context.getResources().updateConfiguration(config,
        context.getResources().getDisplayMetrics());
}

public static void saveLanguage(Context context, String lang) {
    context.getSharedPreferences("APP_LANG", Context.MODE_PRIVATE)
        .edit()
        .putString("lang", lang)
        .apply();
    setLocale(context, lang);
}
}
```

3. Adding New Crop (Firestore Upload)

```
Map<String, Object> crop = new HashMap<>();
crop.put("name", cropName);
crop.put("quantityKg", qty);
crop.put("pricePerKg", price);
crop.put("location", location);
crop.put("harvestDate", harvestTimestamp);
```

```
crop.put("biddingEndDate", endTimestamp);
crop.put("isSold", false);
crop.put("farmerId", FirebaseAuth.getInstance().getUid());
db.collection("Crops")
    .add(crop)
    .addOnSuccessListener(ref -> Toast.makeText(this, "Crop Added!",
Toast.LENGTH_SHORT).show());
```

4. Real-Time Crop Loading (Customer Dashboard)

```
Query q = db.collection("Crops")
    .whereEqualTo("isSold", false)
    .orderBy("biddingEndDate", Query.Direction.ASCENDING);

cropsListener = q.addSnapshotListener((snapshots, e) -> {
    cropList.clear();
    for (DocumentSnapshot ds : snapshots.getDocuments()) {
        Crop c = ds.toObject(Crop.class);
        c.setId(ds.getId());
        cropList.add(c);
    }
    adapter.notifyDataSetChanged();
});
```

5. Placing a Bid (Customer Side)

```
Bid bid = new Bid(userId, name, bidValue, new Timestamp(new Date()));

db.collection("Crops")
    .document(cropId)
    .collection("Bids")
    .add(bid)
    .addOnSuccessListener(r -> Toast.makeText(this, "Bid placed!",
Toast.LENGTH_SHORT).show());
```